AeroSense mechanical design and evidence boundaries

Design freeze: 2026-10-05. This report introduces the mechanical decisions and then includes the complete optical and airflow analyses. This is a traceable engineering-prototype package, not certification of physical fit, cell compatibility or odor recognition. The PCB has not been manufactured; no fully populated STEP model was available. A guaranteed first-print fit would therefore be an unsupported claim.

Current operating conclusion

The eight STL files are unchanged. The fixed-geometry flow sweep selects 3 mL/min as a conditional starting point: nominal worst gas t90=50.00 s; finer-grid check=50.98 s; half-timestep=49.85 s. This uses a provisional 60 s / 90% inlet-concentration target, not a known biological threshold. Use one externally regulated downstream pump and physically verify flow; software settings do not measure or regulate flow. The protein-assisted capture mechanism remains unvalidated.

The UI now supports measured illuminated-blank I/Q subtraction, a measured fluorescence phase reference, unsmoothed corrected export and timestamp-based display EMA. Raw data remain intact. Constant subtraction cannot remove changing same-frequency contamination or shot noise. One cell-free standard-liquid experiment tests hardware response and optical transfer; it does not establish odor sensing.

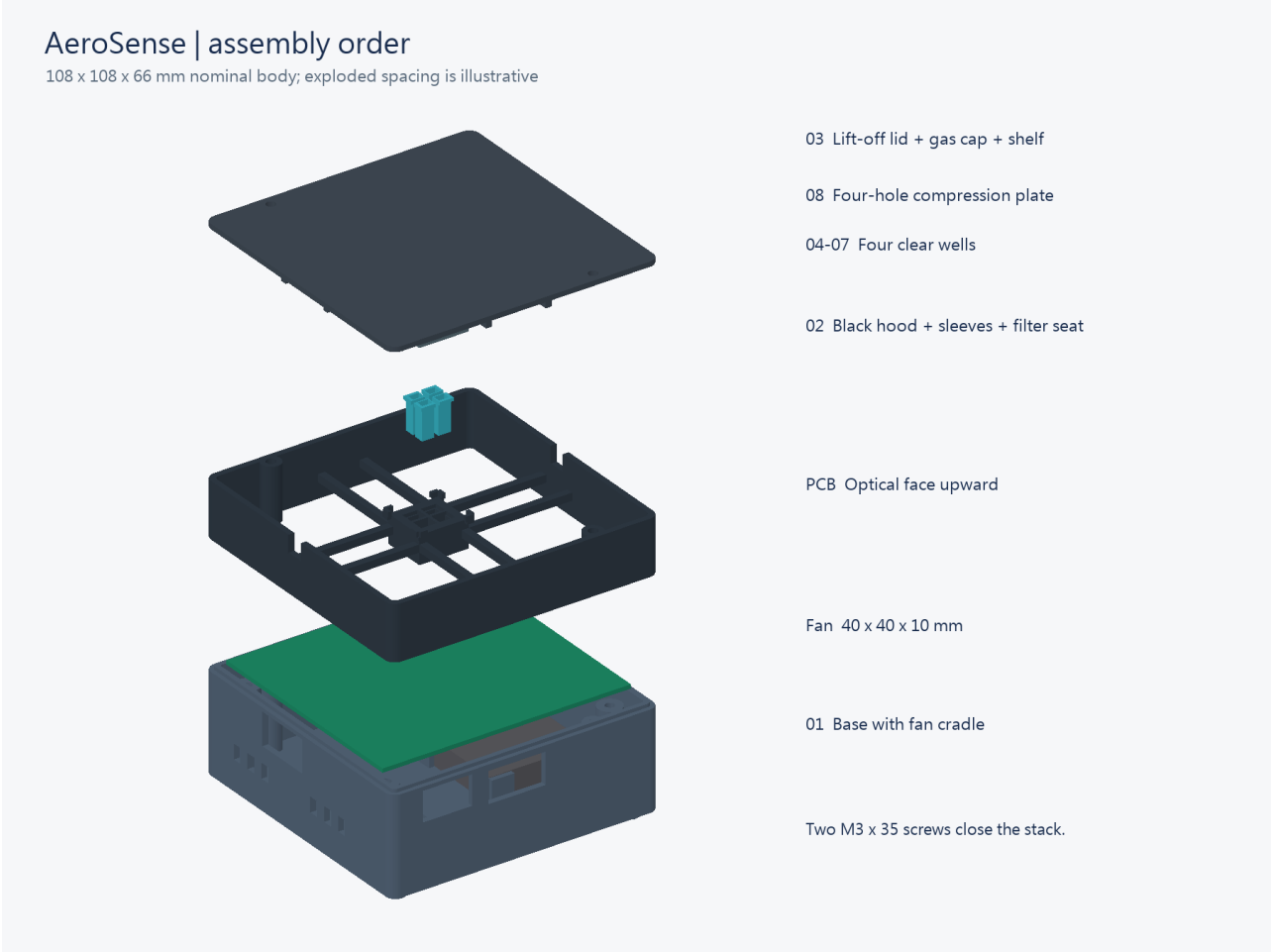
The later liquid/PDMS chapter is a conditional no-protein/no-mixing sensitivity, not an observed response-time result.

Design decision

The user confirms bottom-adherent cells and100 µL per well. The transport analysis therefore includes the approximately8.93 mm liquid column and evaluates a PDMS film at the well mouth. Retain the eight-part experimental carrier and do not add PDMS to the selected assembly: hydrophobicity does not establish a water-vapor barrier, and target-specific membrane properties are unknown. Gas arrival at the mouth is not exposure at bottom cells. With stagnant liquid, assumed D=10□□ m²/s and an immediately fixed surface concentration, bottom t90 is approximately22.83 h. This is a conditional transport warning, not a measured biological response time. Assumptions, sweeps and limits are documented in the later PDMS/liquid chapter.

The nominal body is108×108×66 mm, excluding external tubing and cables. Its bounding volume is31.9% below the previous114×109×91 mm envelope; this does not establish measured mass or battery endurance. There are eight prints: four black structural parts and four independent flat-bottom clear wells. Two long screws release the lid. A separate hinge, fan bracket, four-valve set and status LED have been avoided; the phone reports status.

Each well is a single piece removed upward. Opaque sleeves, a minimum1.2 mm shared wall, the filter seat and support arm are integrated into the hood. Only the Z42.5-44.2 mm bottom optical band remains open. The well must not press directly onto LED or PD packages. Wells follow the LED positions in four quadrants; their centre vectors are not exactly90° apart. The provisional PD sensitive-centre offset is(0,□0.4) mm, distinct from the package centre.



Assembly exploded view rendered from the actual STLs

The separate four-hole compression plate distributes lid load over all well seals. Removing it merely to reduce part count would leave the inner well mouths inadequately supported. The plate can be removed for cleaning; the integrated gas shelf has an accessible underside instead of an enclosed perforated cavity.

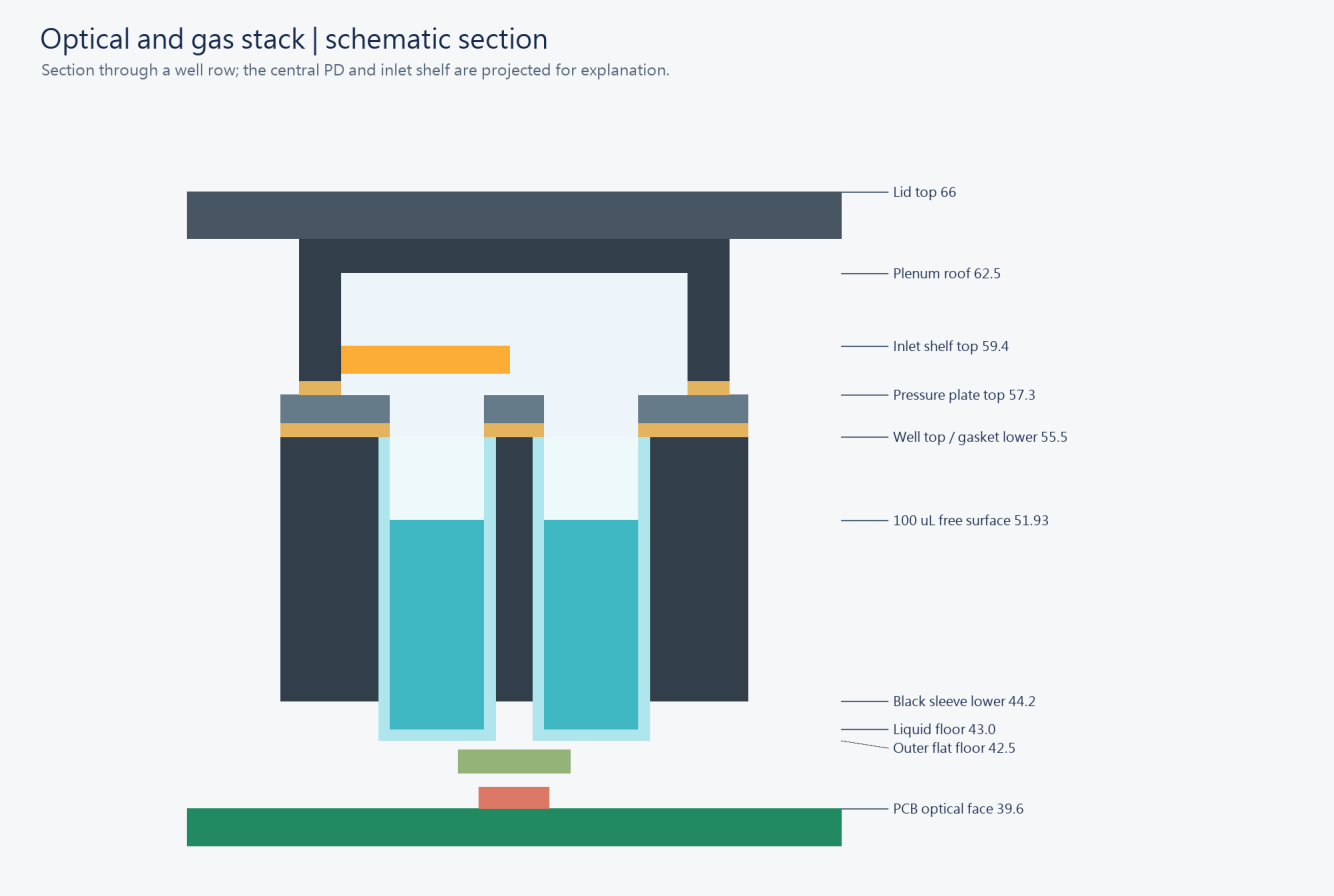
Dimensions and allowances

Item	Design value and boundary
PCB	100×90×1.6 mm; optical face Z39.6. Thickness assumed; four mounting centres±45/±40 mm inferred from PDFs
Fan	40×40×10 mm,5 V,32 mm pitch; integral cradle,0.3 mm clearance per side; two diagonal M3×16 screws
Under-board space	22 mm above fan,18.5 mm above screw heads to bare PCB; battery holder, battery and underside parts unverified
USB-C	Front negative-Y opening: X31.5 centre,20 mm wide, Z28.5-40.5; generous PDF-based service opening, not a measured connector fit
SW1/service	Right-side23×13 mm switch access; bottom J3 and SW2/SW3 access. Bottom-view mirroring must be confirmed
Clear well	5×3.8 outside,4×2.8 inside,0.5 mm flat floor;12.5 mm inner height,140 µL brim,100 µL working fill
Well sleeve	5.3×4.1 slot,0.15 mm nominal clearance per side; shoulder-supported flange retained by sealing, no brittle resin snap flexure
C21 relief	0.5 mm nominal vertical clearance above the assumed1.2 mm component envelope; height unconfirmed
Filter/seals	Maximum filter5.2×4.2×1 mm; three gasket shapes cut from0.8 mm sheet, nominally compressed to0.6 mm
Gas chamber	14.8×12 mm fluid plan, Z57.3-62.5;2.4 mm bore; selected shelf7.2×8×1.2 mm

Geometric volume is4×2.8×12.5=140 mm³=140 µL. At100 µL the liquid depth is8.93 mm and headroom to the mouth is3.57 mm. Meniscus, printing error and cells are excluded.

Nominal gasket compression is(0.8−0.6)/0.8=25%. This is a geometric preload target, not a leak rating. Hardness, tolerances, warping and screw loads determine real contact. Do not force the lid to overcome interference. No tightening torque is validated; close progressively and evaluate a cell-free leak test.

The0.15 mm sliding allowance is not a16K-printer accuracy specification. Exposure, shrinkage, orientation, washing and post-curing can exceed it. Print hood02 and well04 first; measure the slot and well and check their fit before completing the other six parts. No unmeasured shrinkage compensation has been applied.



Optical and gas sealing stack

Interfaces and purchases

According to the schematic, J2 USB-C supplies power/charging and is not an ESP32 programming interface. Program through J3 with BOOT/RESET as detailed in the operation guide. J3 pin1 is3V3, pin2 TX, pin3 RX, pin4 GND. When the PCB is powered normally, do not connect UART-adapter VCC; cross TX/RX and use3.3 V logic only. Confirm any orientation ambiguity with the vendor.

Mechanical purchases are one fan, eight M3 screws, eight nuts, six washers, one silicone sheet for three seals, and short tubing/couplings. Inlet/outlet gas filters and a fluorescence filter remain functional requirements. Gas-filter pressure drop and odor adsorption are not included in the chamber model. Confirm fan current and its supply connection; a GPIO is not a fan power output.

The very small fluorescence filter is a critical procurement dependency: a suitably sized part with suitable spectra and oblique-incidence behaviour has not been selected. Confirm availability before batch-printing the optical hood. Mechanical openings do not create electronic interfaces that the PCB lacks.

Materials, cooling and contamination

Use PC/GF-like Black for all four black structures, including fit checks. Measure transmission through the actual black hood and check seam leakage after full assembly. Use Aqua Clear Plus wells initially for dimensions, water-leak and cell-free optical tests. Ideal simulations omit its tint, absorption, autofluorescence and roughness. Polishing changes a surface and cannot remove intrinsic spectral absorption. Direct HEK293T contact, residues and sterilization are unqualified; complete post-curing is not proof of cell-culture compatibility.

The fan sits under the PCB, separate from the well-mouth gas path. This CFD models the gas cap and well headspaces, not fan cooling, heat generation, battery temperature or37°C control. No cell-temperature

sensor or thermal feedback loop is provided. Reliable thermal, battery-life and weight claims requiremeasured power, temperature, airflow and mass.

Liquid is separated; gas is shared. The wells are not independent sterile gas samples. Seals may reduce leakage and spills but neither sterility nor pressure integrity is validated. Stop inlet flow and keep the outlet vented while closing; resume at low controlled flow only after closing. Do not clamp both ports and compress the lid. External filters can dominate the approximately0.009 Pa simulated internal pressure drop. That result does not establish harmless lid-closure pressure or cell protection.

Executed checks and limits

All eight operating STLs are single connected, closed, positive-volume meshes. Printed-part intersections and nominal bare-PCB, fan, PD/C21/R27/U8 envelopes show no positive-volume interference. This is a check of those specified proxies, not all populated components. Details are in 09_engineering_sources/CAD/checks. Resin stiffness, fatigue, barb strength and drop performance were not experimentally tested; no strength or ruggedness rating is claimed.

Optics and CFD use the final optical core, compression plate, sealing stack and selected shelf. Both use explicitly defined geometric representations, not full triangulated-STL multiphysics. The following analyses document equations, sweeps, errors and limitations. Optical transport supports a testable collection path, not adequate biological signal. The airflow choice is a conditional tradeoff, not a global optimum or cell-protection proof.

Firmware and LittleFS compile successfully. Software, normalization and DLIA checks use synthetic inputs. BOARD_VALIDATED defaults to false for dark-only acquisition until bench checks pass. The phone schedules/stops acquisition, displays frequency amplitudes, calibrates and exports. Missing temperature, battery, flow and trained-classifier data are explicitly identified rather than fabricated.

Shortest path to a physical decision

1. Read the assembly manual and 04_manuals/Vendor_Confirmation_EN.md. Obtain populated STEP, thickness, hole diameters, interface positions, battery envelope, optical heights and PD rotation. The request is drafted, not sent.

2. If printing immediately, check dimensions and fit with hood02 and well04. Take the eight operating files only from 01_print. Assembly-coordinate files are not supplied separately; engineers can regenerate them from the CAD source when needed.

3. Follow the ten-day plan after PCB arrival: electronics/frequencies; cell-free three-level and single-well matrix tests; fit/light-leak/water checks; then3-5 manual users and two retests.

4. Record and stop on interference, ADC rail clipping, incorrect single-LED timing or leakage. Do not hide hardware defects with normalization. Consider a small biological confirmation only within an already established laboratory workflow after the four core tests.

Cost and iGEM evidence

The BOM separately lists prints, the vendor's61 PCB rows, and purchases. The NT$131,250 quote covers two boards including design/setup; NT$65,625 is the allocation per prototype board, not a production quote. Known non-PCB material consumption is approximatelyNT$768.32-1,041.32, with seven required purchase rows and four shared-equipment rows still unquoted. It is not a complete cost. Resin use is geometry×density×support/waste assumptions, not measured consumption. Yellow spreadsheet cells accept actual prices; PCB components are not double-counted on top of the assembled-board quote.

Hardware evidence should connect a synthetic-biology need, user input, a functioning proof of concept, and reproducible documentation. The06_wiki package separates these evidence types. Cycle4 documents the enclosure/interface/shielding/flow/seal redesign and calculations. Cycle5 currently documents a study plan; observed improvements must await real testing. See the officialSpecial Prizescriteria.

The critical missing evidence is actual board geometry, blank leakage, measurable fluorescence and repeatable reassembly. Additional simulations cannot replace those measurements.

Optical-path and frequency analysis

Design decision

The new opaque carrier has nonzero optical collection and is suitable for a cell-free optical prototype. It has not demonstrated complete well isolation, uniform illumination or sufficient biological detection sensitivity. Wells are registered to the LEDs. The modeled PD-sensitive center is offset (0, □0.4) mm from the package datum; assembly rotation and bottom-view mirroring still require vendor confirmation.

The carrier encloses the clear walls from Z 44.2 to 55.5 mm, with a minimum central opaque web of 1.2 mm. The bottom band Z 42.5-44.2 remains open for excitation and collection. That shared band preserves the optical path but also permits cross-well excitation. Actual black resin opacity at 470/520 nm has not been measured. Retain the carrier, flat-bottom wells and PD seal; test filtering and reseating repeatability before cell work. Do not remove the seal merely because the predicted signal increases.

Geometry and assumptions

Item	Model
LED centers	(±3.3,±5.7)mm; side emitters facing central Y direction
Emitting area	1.0×0.3 mm, Z 40.2 mm; Lambertian, no package lens
Well centers	(±3.3,±2.65)mm
Outer/cavity dimensions	5×3.8 mm /4×2.8 mm; outer floor 42.5, inner floor 43, top 55.5 mm
Fill	100 µL per well; flat meniscus Z 51.93 mm
Opaque carrier	20×18 mm outside; four 5.3×4.1 mm clearance pockets
Four-well compression plate	20×18 mm,Z 56.1-57.3, four 4×2.8 mm holes; four-hole lower gasket Z 55.5-56.1
Common cap	Upper perimeter seal Z 57.3-57.9,14.8×12 mm opening; cap outside 18.4×15.6 mm, lower 57.9, inside roof 62.5, outside roof 64 mm
Inlet shelf	X□7.4..□0.2,Y±4,Z 58.2..59.4 mm, represented as an opaque solid
PD	Top Z 40.5 mm;3×2.5 mm equal-area 7.5 mm² rectangular proxy, not measured die shape
PD seal	Outer 4×4 mm, aperture 3×3.4 mm, compressed Z 40.5-41.1 mm
Filter	5.2×4.2×1 mm, Z 41.1-42.1 mm; homogeneous absorptive proxy
Refractive indices	Air 1.0, liquid 1.333, clear material 1.5; Fresnel/TIR retained
Ideal clear material	Zero parasitic absorption and scattering
Cell layer	20 µm at the floor; excitation absorption 0.5/mm, assumed
Representative wavelengths	Excitation 470 nm/emission 520 nm; no spectral integration
Reference radiant power	LED-on 500 µW, QY 1 upper-bound scenario; not measured LED output

GCaMP calcium dynamics change the fluorescence envelope; the cell does not have to perform 41-113 calcium responses per second to follow optical excitation modulation. The original GCaMP6 study does not supply this project's expression, cell density, odor response or phototoxicity threshold. Chen et al., Nature 2013

The PCB plane at Z 39.6 is a lower escape/loss boundary; substrate back-reflection is not modeled. The black-reflectance sweep applies to the represented enclosure and component solids, not measured solder-mask, metal-pad or LED-package reflectance. The real empty-well leakage test therefore remains essential.

This is rectilinear material-boundary transport, not arbitrary STL triangle tracing. Main wells including top flanges, carrier including flange pockets, compression plate, inlet shelf, PD, filter and nearby component envelopes are represented. The filter arm is X□8.35..□3.05/Y±0.3/Z 40.9..42.1; its column is X□9.5..□7.5/Y±1/Z 40.9..50.9. The underside relief above C21 is included. Remote enclosure walls and structural bridges use rectangular proxies; rounded corners and external service apertures are treated opaque. Curved lenses, chamfers, roughness and ambient light through external gas ports are not fully represented. A black outer model boundary cannot certify a light-tight product.

Computation and checks

The primary sweep contains 24 conditions×3 seeds, 24,000 excitation and 60,000 emission packets per run: 6,048,000 packets. Cell absorption events seed isotropic fluorescence with appropriate absorption weights. Power conversion is P_LED×absorbed_fraction×QY×470/520×collection_fraction.

Additional maps use 600,000 excitation packets and 224 isotropic source locations×65,536 packets. Collection means the fraction of source power reaching the PD, not peak-normalized brightness. Per-pixel SE and 95% Wilson intervals are saved. Median relative SE is 17.3%, with 0 zero-hit pixels. A zero-hit estimate is not proof of zero collection.

Sweep bars are mean±1 SE across three seeds, not 95% confidence intervals, and exclude material/biological/model uncertainty. With only three seeds, small rankings should not be interpreted as optimization proof.

Analytic rectangular solid angle, 4% normal-incidence Fresnel reflection, TIR, HG mean cosine and 10,000 random material-classification points passed. Maximum energy-ledger error was 2.9e-15. Primary runs cap transport at 2500 events; maximum unfinished fraction was 0.0117%, with a worst conservative unresolved green-power bound of approximately 0.121 nW in an individual run. Three additional higher-packet NE comparisons at 2500/10000 events are saved in event_limit_convergence.json. This check addresses the reference condition, not full convergence for every case.

Power and uniformity

NE reference green power is 0.593±0.035 nW; NW/SW/SE are 0.723/0.829/1.008 nW. These are LED-on powers. At 25% duty, time-average power is 0.25 times the on-state power and the sinusoidal fundamental amplitude is approximately 0.450 times it. On-state power must not be substituted directly for lock-in amplitude.

Mean upward NE floor irradiance is 4.366 μW/mm², with spatial pixel CV 46.2%. This is upward crossing flux per floor area, not all-direction fluence. It excludes cellular distribution variation. A flat floor does not imply uniform illumination. Side illumination, component shadows and refractive interfaces still create spatial structure.

Reversing PD offset gives 0.994 nW; centering it gives 0.922 nW. Removing the seal gives 1.103 nW, but sacrifices the light/sealing boundary and is not recommended.

Only 19.3% of collected reference green power is within 45° of the normal and 64.8% within 65°. These are artificial acceptance-cone diagnostics. The VEMD5060X01's±65° specification is a half-sensitivity angle, not a hard cutoff. The model does not multiply an additional cosθ factor because finite-area geometric projection is already included. The actual package response and oblique-incidence filter spectrum remain uncalibrated. Vishay datasheet

Cross-well transfer and reflection

Active LED /emitting well	NE	NW	SW	SE
NE	0.5260	0.0506	0.0000	0.0168
NW	0.0624	0.6571	0.0031	0.0000
SW	0.0000	0.0000	0.7409	0.0881
SE	0.0201	0.0000	0.0943	0.8933

Units: nW. Each row separates received fluorescence by its well of origin. Intended-well fractions are 88.6-90.9%; approximately 9.1-11.4% comes from other wells. Finite-sampling zeros do not establish perfect isolation. The simulated matrix condition number is 1.86. This suggests possible numerical separation under these assumptions, but the actual 4×4 matrix must be measured using a single fluorescent standard well at a time. Fill, cell distribution and reseating can change it.

Approximately 32.0% of received NE green power underwent at least one Fresnel reflection after fluorescence emission. Reflection is not automatically interference: reflected target-well fluorescence can still be useful signal. This counter does not combine the earlier excitation-path history with the emission history.

Black reflectance 0.02/0.10 gives 0.877/1.303 nW; liquid scattering 0.1/1.0 mm□¹ gives 0.813/2.122 nW. These are sensitivity cases, not recommendations to make wells reflective or cloudy. Higher total collection does not establish better channel independence.

At liquid scattering 1/mm, the intended-well fraction is 84.3%. Removing the upper carrier gives an NE intended fraction of 90.75%, versus 88.63% with the carrier. These comparisons must be read with finite-sampling uncertainty. The sleeve does not eliminate shared-bottom cross-talk. The carrier provides positioning and external-wall shielding; actual ambient-light rejection needs an empty-well illumination test.

Filtering and synchronous background

Without excitation rejection, received blue power is 110.5 nW, approximately 141 times that case's green power. Absorptive OD 3/4/5 proxies give blue powers 0.02423/0.00149/0.00010 nW. Oblique rays travel farther in this absorptive model. These values must not be applied as real interference-filter angular specifications.

The linear signal is $y(t)=\Sigma[g_j(c)+b_j]u_j(t)+n(t)$. Cell-fluorescence gain g_j and synchronous background b_j both retain their LED's modulation. Reflection changes path/weight; it does not automatically turn 41 Hz into another frequency. Lock-in extraction suppresses unrelated-frequency noise but cannot guarantee removal of reflected excitation light.

The detector comparison now uses a separate three-seed MC calculation for each LED at OD3, OD4 and OD5: 12 additional cases and 3,024,000 packets. The original 24-case results are unchanged. Each LED has its own blue leakage; no NE gain is reused as another LED's prediction. The desired green comparator uses the OD3 green means (emission normal transmittance 0.9), retains reflected fluorescence, and stays fixed across the synthetic rejection sweep.

LED	Desired green on-state nW	Blue optical on-state nW	Blue MC SE nW	Amplitude bias %	Error-inclusive SINR dB
NE	0.65060	0.02423	0.00423	2.781	31.12
NW	0.55182	0.01634	0.00647	2.626	31.61
SW	0.70940	0.02934	0.00515	3.283	29.67
SE	0.83776	0.03769	0.00292	3.521	29.07

All four frequency signals use 4 kHz sampling, two seconds and 25% duty. The default assumes R470/R520=0.8 and R520=0.3 A/W; these are not measured detector specifications. Plots use green-equivalent nW, not measured current or voltage. Ambient is 0.15 nW DC plus a 0.05 nW cosine at 100.3 Hz, phase 0.4 rad. Independent photoelectron counts obey Poisson statistics at the total fluorescence + leakage + ambient rate; electronic white noise has 0.005 nW per-sample RMS. The gate is held at its point-sampled state during each 250 µs bin; true PWM-edge overlap, TIA bandwidth, ADC response, dark current and jitter are omitted.

The ideal comparator removes parasitic excitation, ambient and both noise sources while preserving the same green trajectory. The residual therefore is not “all reflected light.” Default waveform RMSE is 0.17631 nW; DC residual 0.17155 nW; AC residual RMS 0.04067 nW; waveform AC SINR 23.40 dB. Default shot/electronic RMS are 0.001358/0.005010 nW. Joint sine/cosine least squares gives complex C; error-inclusive SINR is 20log10(|Cideal|/|Creceived□Cideal|). Noise-only SNR instead uses repeat-to-repeat fluctuations and excludes systematic bias.

The synthetic sweep has 144 scenarios × 32 repetitions: OD∞/3/4/5, four ambient cases, electronic RMS 0/5/20 pW and responsivity ratios 0.6/0.8/1.0. Its 18,432 channel records and 576 aggregated rows are saved in interference_sweep*.csv. MC means are held fixed, so these repeated seeds do not propagate MC or biological uncertainty. Same-frequency ambient remains even with ideal excitation rejection; a high noise-only SNR can coexist with biased amplitude.

The 20-second MP4/GIF uses 200 common timestamps at 10 fps and 100× slow playback. Every panel starts at sample zero. Cursor, trace endpoint, LED state and path gating share the same sample index. Only actual NE sampled paths are drawn; no mirrored paths are presented as other LEDs' trajectories. The PD active-center cross marks the provisional offset (0,□0.4) mm. Paths switch together, without fabricated travel delays. This does not simulate calcium dynamics or validate hardware timing. Detailed methods, results, axes, legends and limitations for all ten figures and the movie are in Figure_Guide_EN.md and figure_notes/.

Is the intensity sufficient?

Nonzero collection is established in the model; odor detection is not. The relevant quantity is ΔF relative to measured noise. conditional_photon_budget.csv varies LED-on power 0.05-2 mW,QY 0.1-1,andΔF/F 1-50%, including the 25% duty fundamental factor.

At 500 µW,QY 0.6,andΔF/F 10%, predicted lock-in amplitude change is 16.03 pW. If SNR□5 is chosen as an engineering target, repeated lock-in amplitude RMS noise must be below 3.21 pW equivalent input power. This is not a per-ADC-sample noise limit. Without measured responsivity, gain, filtering, baseline drift and GCaMP expression, it is not a defensible detection limit.

Minimal two-week validation

1. Dark and empty-well background: collect ten 2 s blocks at identical settings; check saturation/timing errors and each channel's mean/SD.

2. One standard well, four readouts: put the same fluorescent standard in one well and matched blank in the others; repeat for all wells to obtain the measured 4×4 matrix for that standard. Compare normalized ratios before claiming absolute nW agreement. Preserve raw and blank-subtracted I/Q. Uniform liquid standards should be compared with the volume-source model below, not used directly for cell-layer unmixing.

3. Three reseatings: repeat removal/reassembly at fixed concentration, volume and settings. Record channel means andCV. CV□10% and standard-vs-blank SNR□5 can be initial internal targets; they are not achieved results, official iGEM requirements or clinical standards.

Large synchronous blanks require filtering/light-leak correction, not normalization that conceals faults. Poor reseating repeatability requires a mechanical fix. Proceed to cells after these inexpensive checks. A fluorescent flat film image is an optional spatial-uniformity validation, not a required extra experiment under the two-week schedule.

Forward model for the liquid-standard experiment

A homogeneous fluorescent liquid emits throughout its volume, unlike the 20 μm floor cell layer. An additional four LED×three-seed weak-absorption volume model uses 100 μL and 0.001/mm excitation absorption, transporting another 1,008,000 packets. Row-normalized source-well fractions are:

Active LED /standard well	NE	NW	SW	SE
NE	80.6%	14.2%	0.0%	5.2%
NW	12.5%	86.8%	0.8%	0.0%
SW	0.0%	0.5%	86.7%	12.8%
SE	4.9%	0.0%	12.6%	82.5%

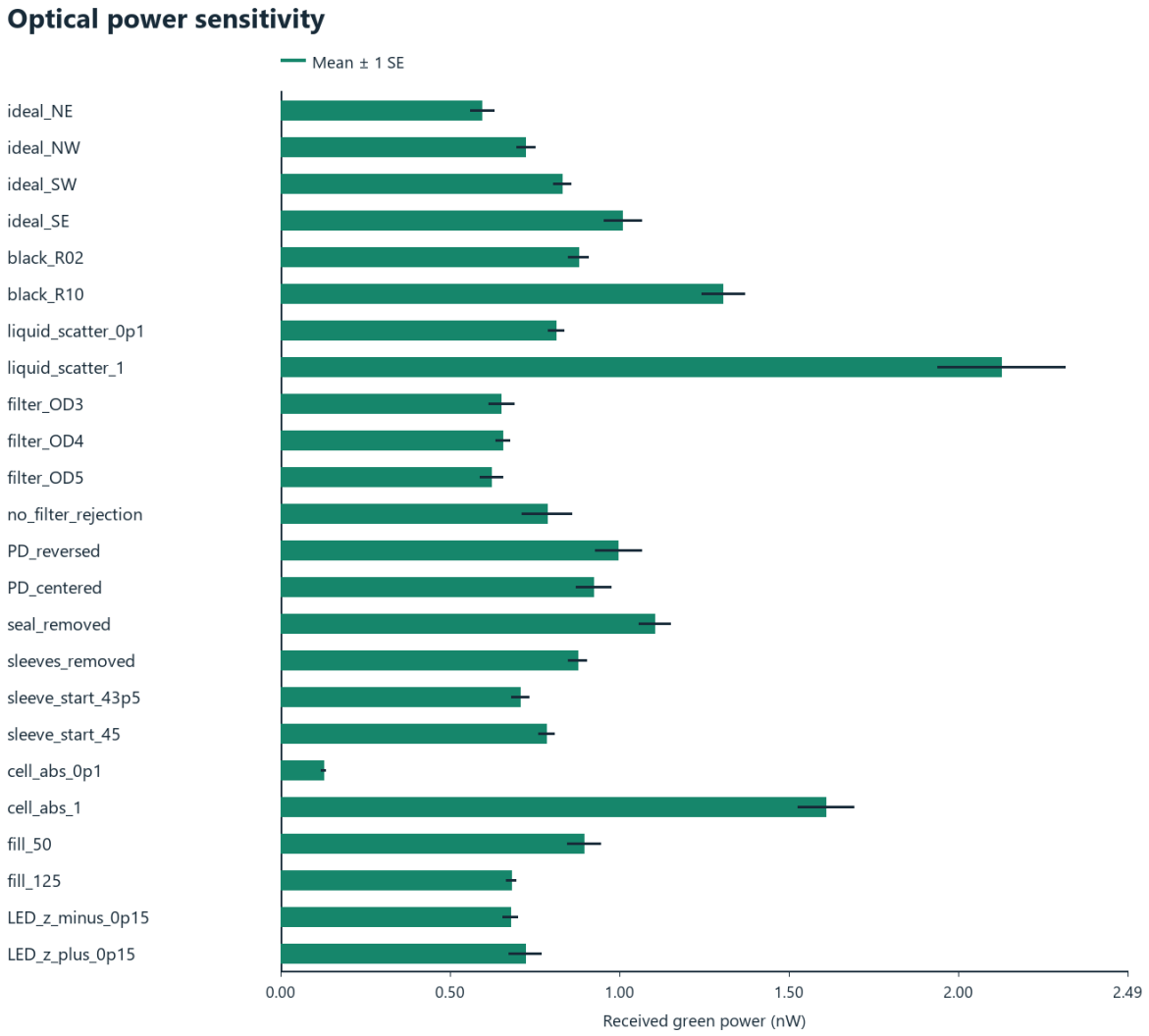
This is a comparison for the same simple standard-liquid test, not an additional required experiment. With unknown dye concentration, spectrum, quantum yield and LED power, compare direction and normalized ratios first, not absolute power. Actual floor-cell unmixing requires a localized bottom standard or independent calibration under cell conditions. Do not substitute a liquid-standard matrix for the cell-layer matrix.

Figures and reproducibility

File	Caption
01_parameter_sweep.png	Conditional received green power for 24 optical cases. Bars are means and error bars are one Monte Carlo standard error across three seeds; they are not confidence intervals for device performance.
02_floor_maps.png	A: upward excitation power transmitted across the resin-liquid floor under NE illumination. B: isotropic point-source power reaching the PD. Each row shares one color scale across all four wells.
03_well_transfer.png	Received green power resolved by active LED and the well where fluorescence originated, for the ideal-material bottom-cell-layer model.
04_detector_angles.png	Cumulative NE green power admitted by artificial detector acceptance cones. Points are three-seed means with ±1 SE.
05_frequency_recovery.png	A: ideal noise-free received fluorescence. B: conditional received signal including MC excitation leakage and assumed ambient, photoelectron shot noise and electronics noise. C: overlay. D: received minus ideal.
06_conditional_budget.png	Maximum lock-in amplitude RMS noise compatible with an engineering SNR target of five, for the stated optical model and quantum yield 0.6.
07_sample_paths.png	A cutaway rendering of actual sampled NE excitation and fluorescence trajectories. Selected detected fluorescence paths are emphasized; line density is not optical intensity.
08_volume_standard_validation.png	Row-normalized source-well contributions for a weakly absorbing uniform fluorescent liquid, distinct from a bottom cell layer.
09_interference_metrics.png	A: ideal/received amplitude spectra. B: signed amplitude bias and complex relative error after joint four-frequency demodulation. C: signal-to-error ratio including systematic contamination and random noise.
10_interference_sweep.png	A/B summarize worst-channel complex error and lowest-channel median SINR at electronic RMS 5 pW/sample and assumed responsivity ratio 0.8. C varies electronic noise; all panels also include assumed photoelectron shot noise.

Detailed per-figure methods and reading instructions: Figure_Guide_EN.md (Figure_Guide_EN.md).

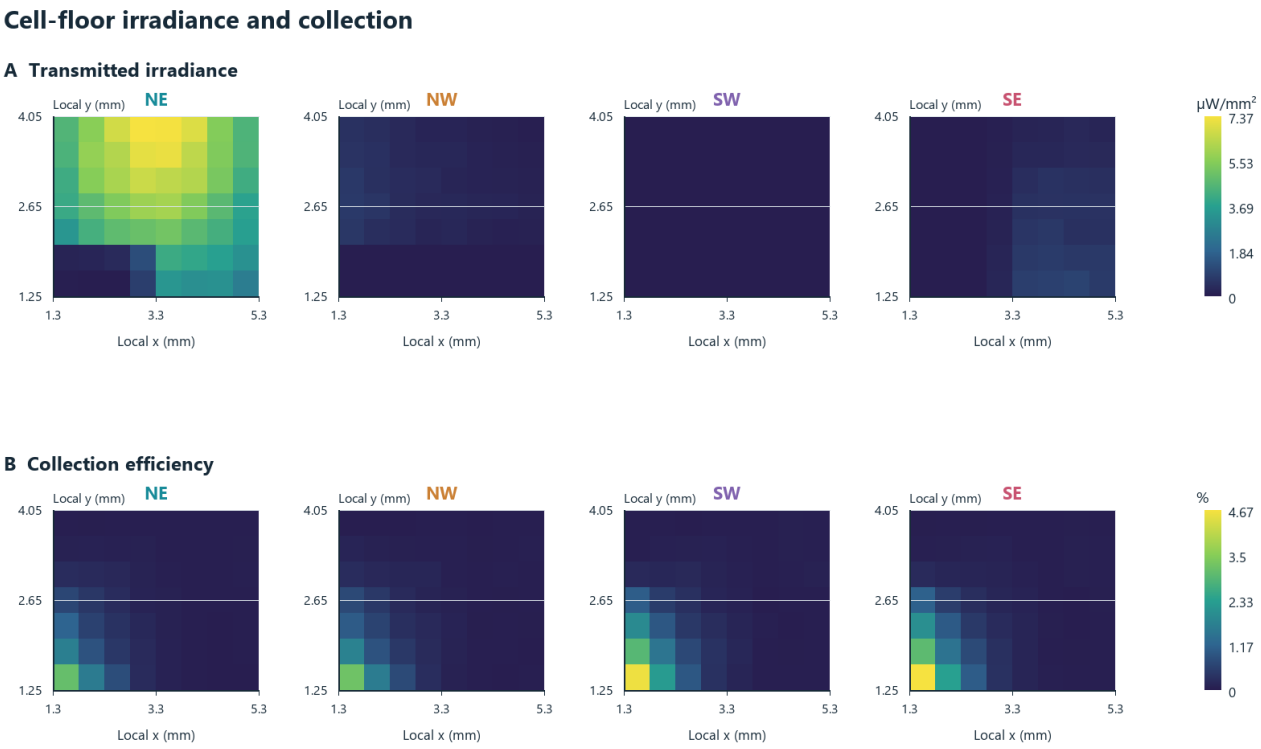
Optical figure / 01_parameter_sweep



Conditional received green power for 24 optical cases. Bars are means and error bars are one Monte Carlo standard error across three seeds; they are not confidence intervals for device performance.

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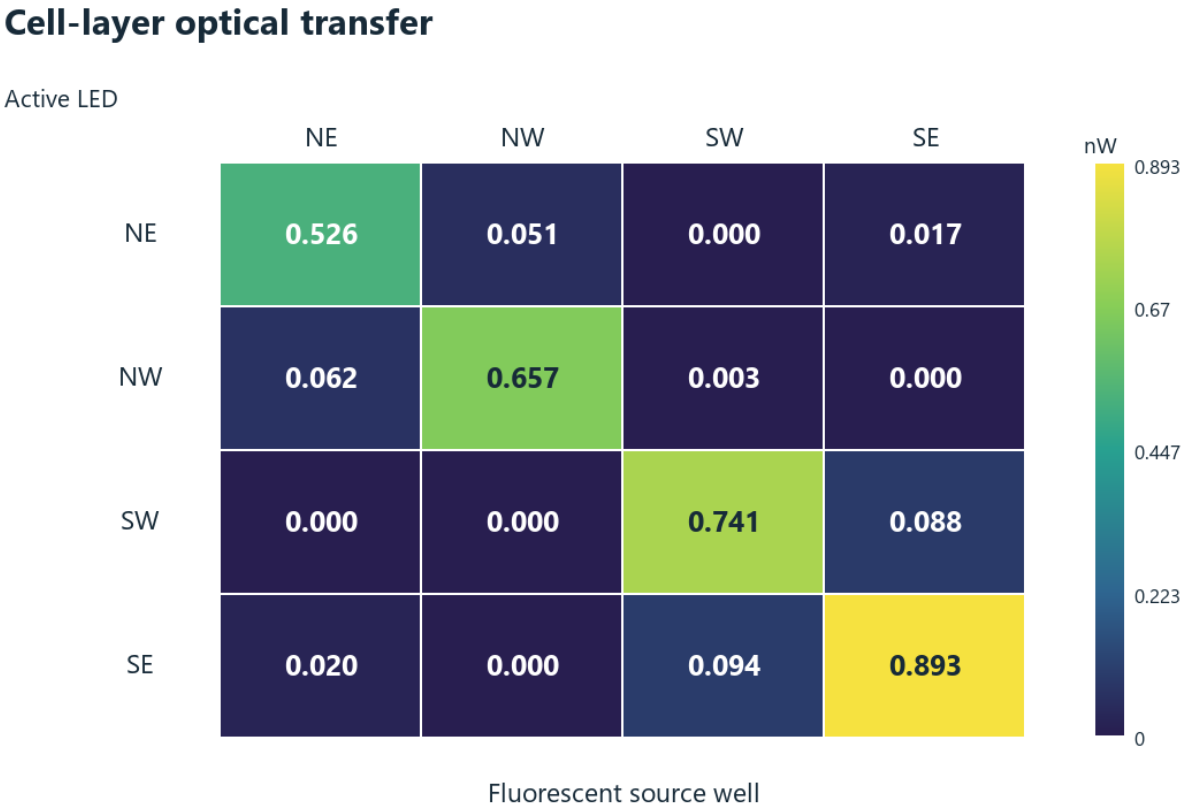
Optical figure / 02_floor_maps



A: upward excitation power transmitted across the resin-liquid floor under NE illumination. B: isotropic point-source power reaching the PD. Each row shares one color scale across all four wells.

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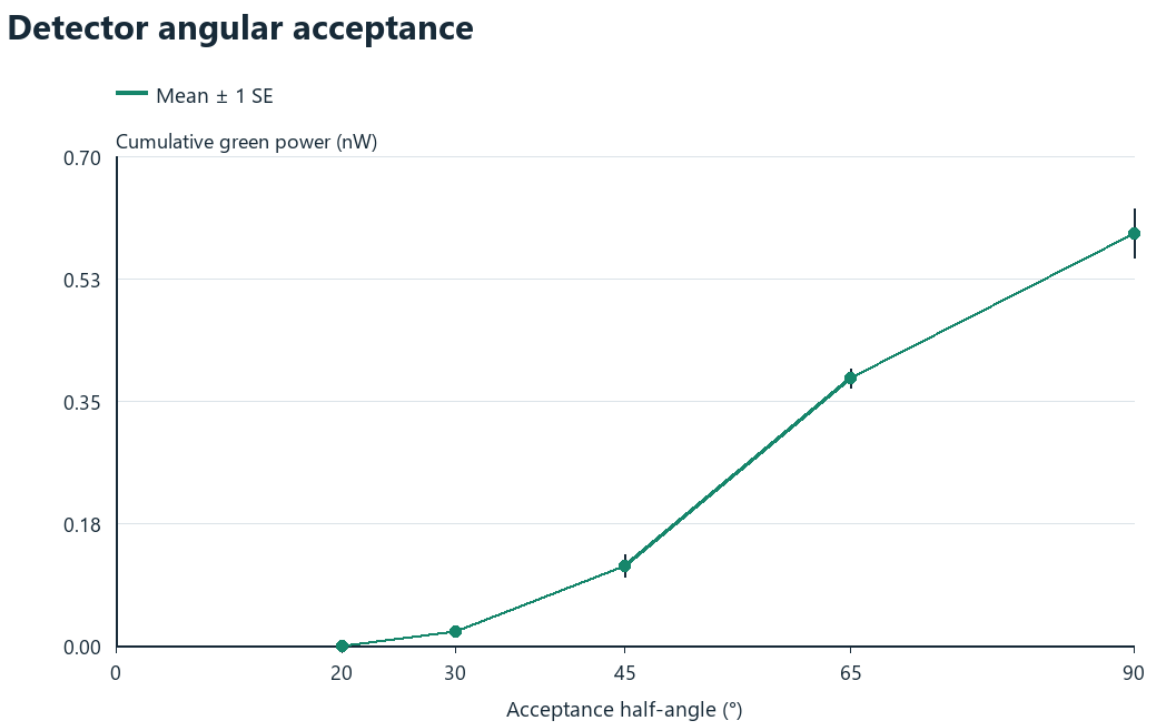
Optical figure / 03_well_transfer



Received green power resolved by active LED and the well where fluorescence originated, for the ideal-material bottom-cell-layer model.

02_simulation\optics\figures\03_well_transfer.png

Optical figure / 04_detector_angles



Cumulative NE green power admitted by artificial detector acceptance cones. Points are three-seed means with ±1 SE.

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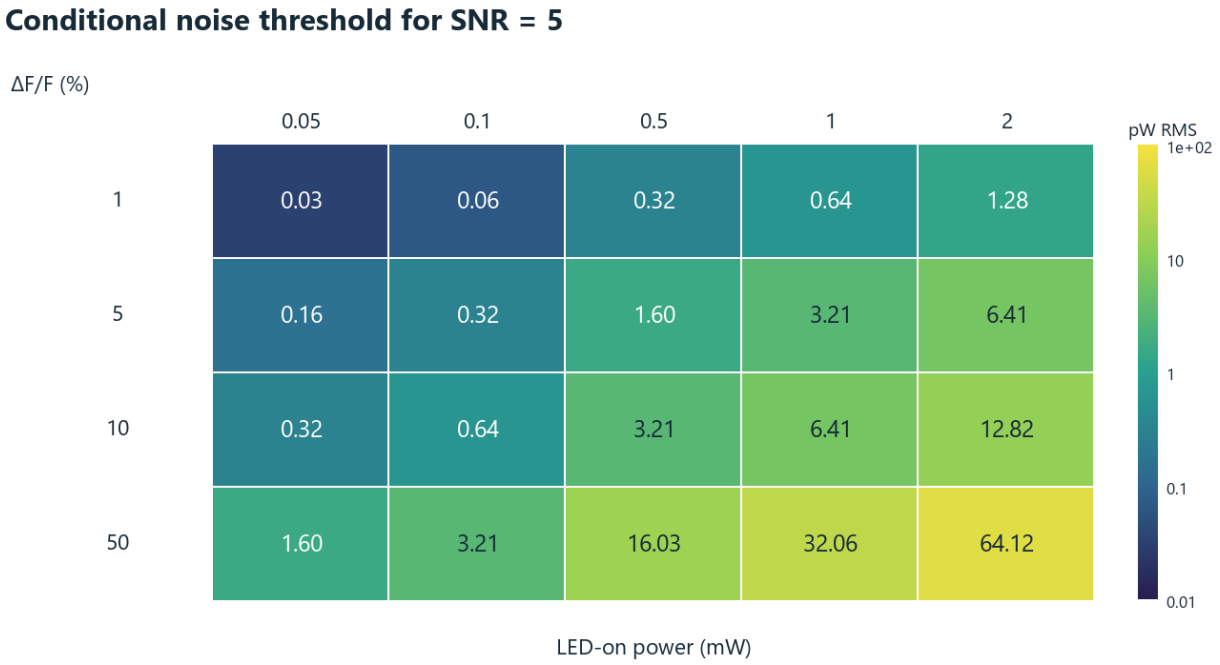
Optical figure / 05_frequency_recovery



A: ideal noise-free received fluorescence. B: conditional received signal including MC excitation leakage and assumed ambient, photoelectron shot noise and electronics noise. C: overlay. D: received minus ideal.

02_simulation\optics\figures\05_frequency_recovery.png

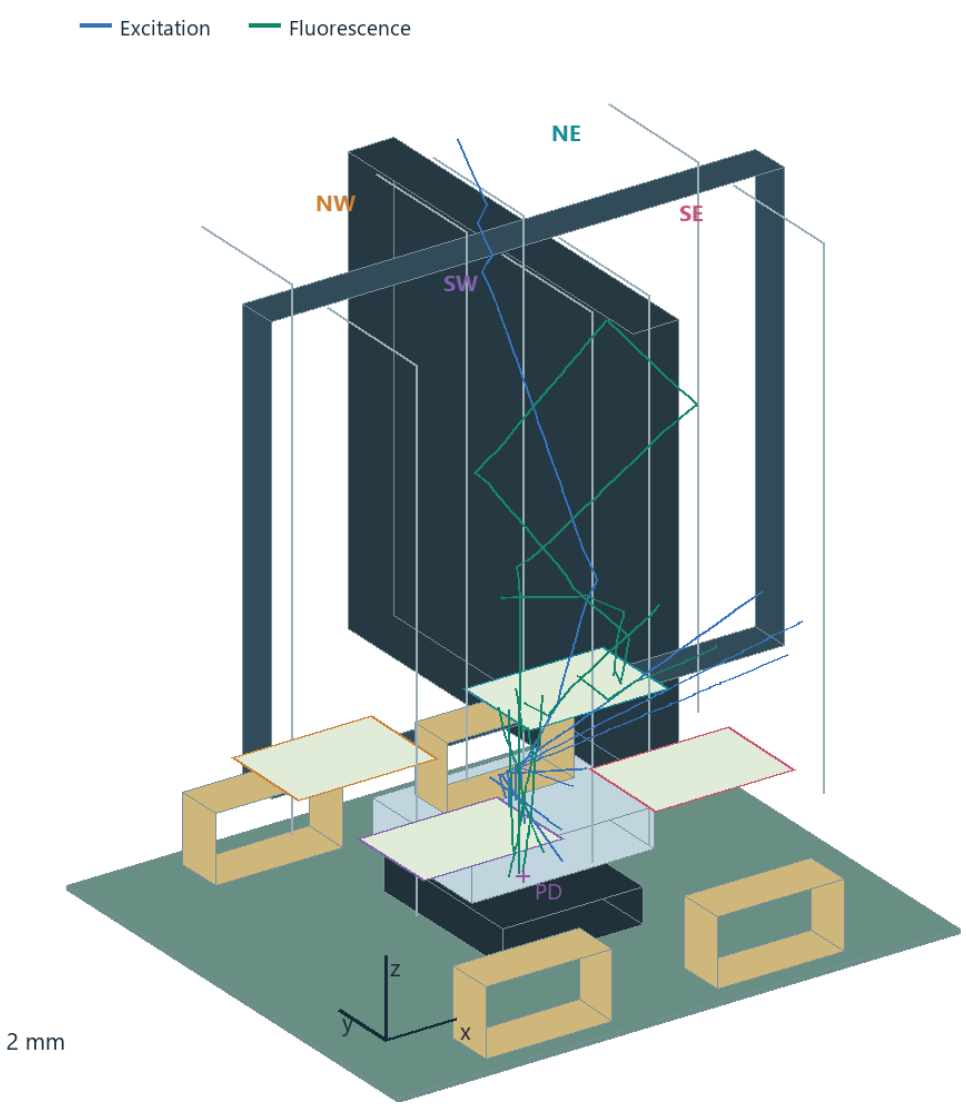
Optical figure / 06_conditional_budget



Maximum lock-in amplitude RMS noise compatible with an engineering SNR target of five, for the stated optical model and quantum yield 0.6.

02_simulation\optics\figures\06_conditional_budget.png

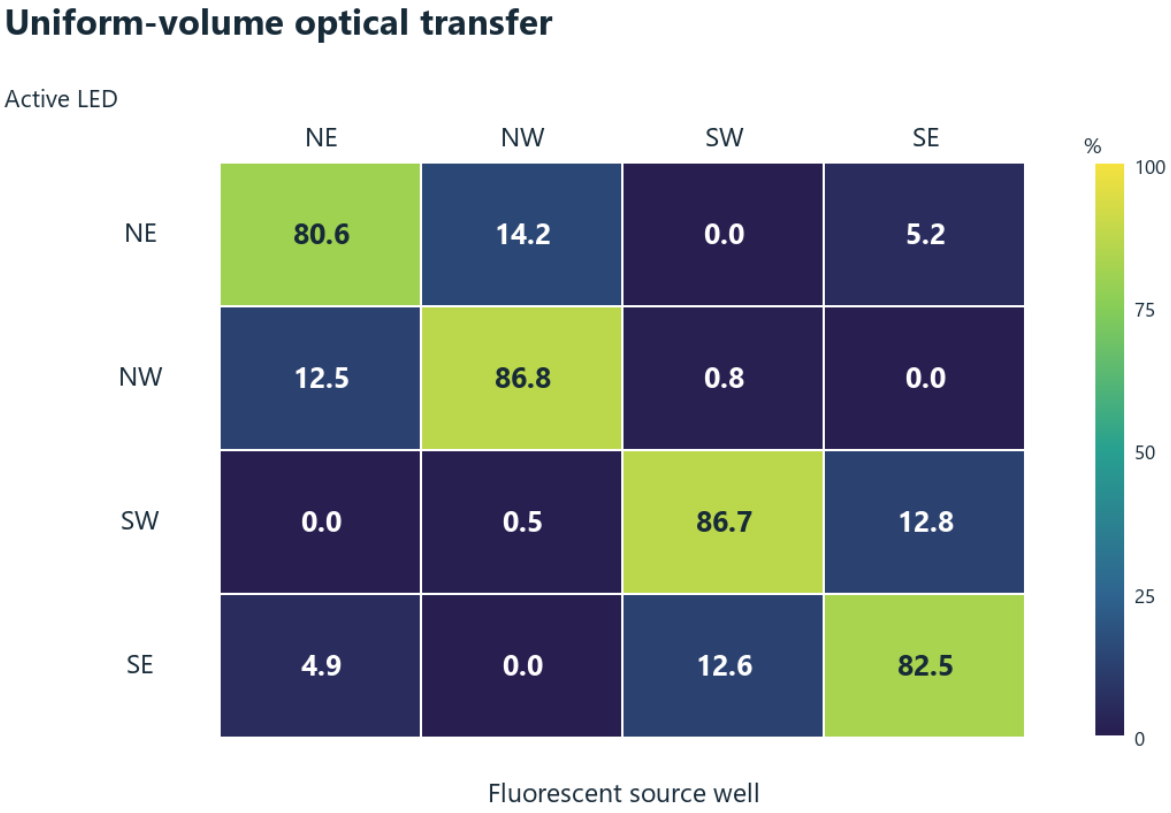
Optical figure / 07_sample_paths



A cutaway rendering of actual sampled NE excitation and fluorescence trajectories. Selected detected fluorescence paths are emphasized; line density is not optical intensity.

02_simulation\optics\figures\07_sample_paths.png

Optical figure / 08_volume_standard_validation



Row-normalized source-well contributions for a weakly absorbing uniform fluorescent liquid, distinct from a bottom cell layer.

02_simulation\optics\figures\08_volume_standard_validation.png

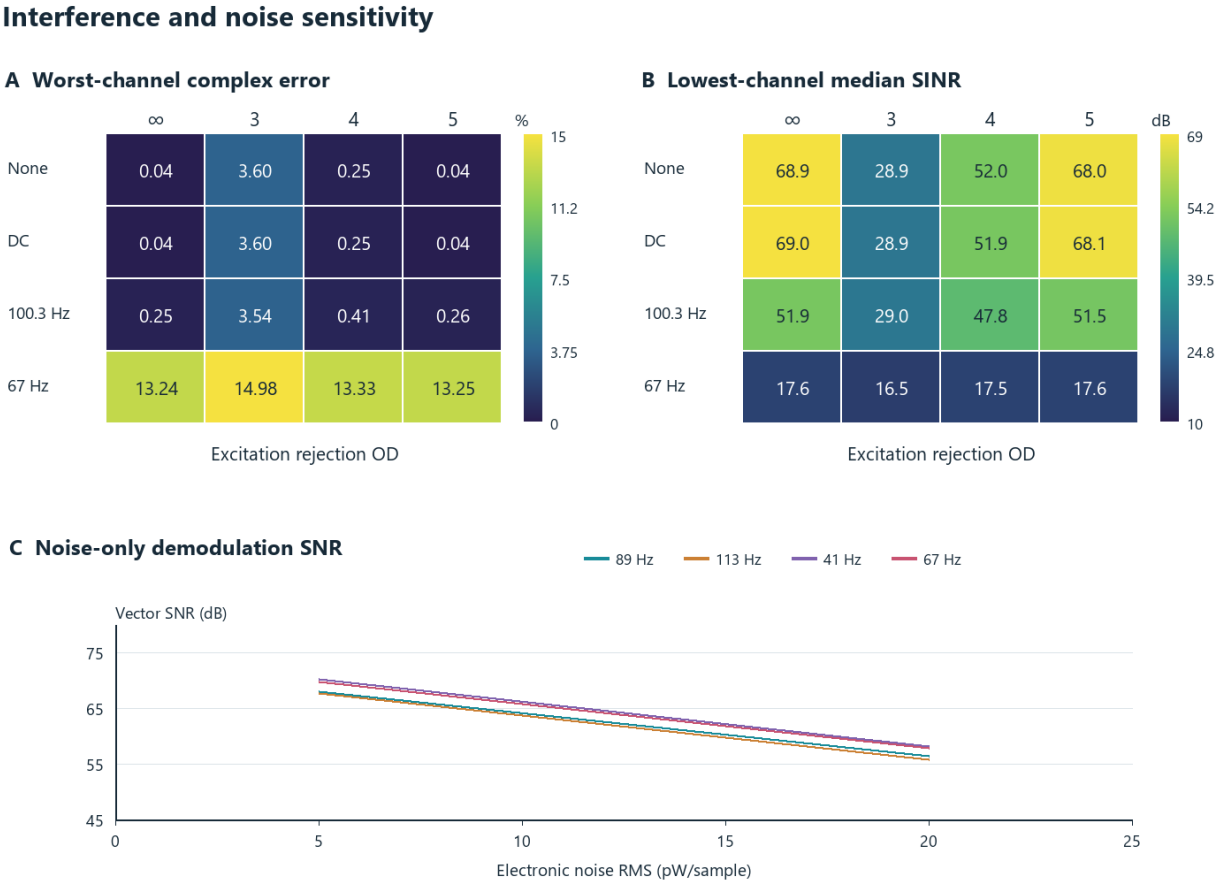
Optical figure / 09_interference_metrics



A: ideal/received amplitude spectra. B: signed amplitude bias and complex relative error after joint four-frequency demodulation. C: signal-to-error ratio including systematic contamination and random noise.

02_simulation\optics\figures\09_interference_metrics.png

Optical figure / 10_interference_sweep



A/B summarize worst-channel complex error and lowest-channel median SINR at electronic RMS 5 pW/sample and assumed responsivity ratio 0.8. C varies electronic noise; all panels also include assumed photoelectron shot noise.

02_simulation\optics\figures\10_interference_sweep.png

Gas operating conditions for the fixed enclosure

The lowest sampled operating candidate satisfying the prespecified conditions is 3 mL/min, with a worst-well/worst-D gas t90 of 50.00 s. This is a conditional choice for the fixed shelf and fixed STL in a gas-side model, not a global optimum, a HEK293T safety result, or a demonstrated time to GCaMP activation. CAD is unchanged.

Prespecified selection rule

Before inspecting the new outputs, operating_sweep_config.json fixed Q=0.25/0.5/1/2/5 mL/min and Dgas=5/10/20 mm²/s. The assumed targets are worst-well/worst-D gas t90□60 s, port Re□3, and near-liquid gas P95 speed□0.01 mm/s. Choose the smallest qualifying Q without a weighted score. These response times assume a sustained inlet concentration step, not a brief odor pulse or cellular response. The 60-s target is provisional; the speed cap is an engineering comparison limit, not a cell-injury threshold. D is an uncertain species/environment property, not a knob to tune like pump flow.

Q (mL/min)	Worst gas t90 (s)	P95 gas speed (mm/s)	Port speed (mm/s)	Port Re	Chamber Δp (Pa)	Meets targets
0.25	549.59	0.000234	0.921	0.147	0.00228	False
0.5	299.09	0.000467	1.842	0.295	0.00456	False
1	149.55	0.000934	3.684	0.589	0.00912	False
2	75.07	0.001869	7.368	1.179	0.01825	False
2.5	60.04	0.002336	9.210	1.474	0.02281	False
3	50.00	0.002803	11.052	1.768	0.02737	True
5	31.95	0.004672	18.421	2.947	0.04562	True

Each t value takes the slowest of four near-liquid gas sampling bands, 0.6-1.4 mm above the liquid; it is not a liquid or cellular concentration. Worst-case D need not be monotonic because upstream diffusion contributes at the prescribed inlet step. V/Q alone is not a response-time calculation. Raw CSV/JSON retain t10, t50, t90 and complete per-well curves. Q2.5 gives about 60.04 s: its 0.04-s nominal miss is smaller than numerical sensitivity, so it is not demonstrated physically inadequate. Q3 gives about 50 s and a larger within-model timing margin; physical testing remains necessary. A rounded figure value of 60.0 is not exact target compliance.

Pareto tradeoff and limits

The nondominated sampled flows are 0.25, 0.5, 1, 2, 2.5, 3, 5 mL/min: lower gas speed generally costs delivery time. Connecting lines join discrete cases and do not certify interpolated operating regions. Port Re approaches 3 at the upper endpoint, so linear low-Re scaling requires checks. Tubing, filters, leaks, adsorption, dissolution and cell response are absent. With 100 µL, the liquid depth is 8.93 mm; delivery to bottom-adherent cells remains a separate liquid/membrane problem (../membrane/Analysis_EN.md).

Numerical method and checks

The main grid is 0.4 mm. Saved rigid-wall/no-slip-liquid gas velocity is linearly scaled with Q; projected face fluxes enforce discrete conservation. Conservative finite volume uses upwind advection, central diffusion and backward Euler. Initially C=0; the inlet terminal steps to C/Cin=1; the outlet is convective with zero diffusion; solid and liquid walls are impermeable. Timesteps are 1 s for Q□0.5, 0.5 s for Q1/2 and 0.25 s for Q□2.5. End time is max(150,300/Q) s.

There are 15 initial operating scalar solves and 6 adaptive bracket-refinement solves at Q2.5/3, declared in a separate stage-2 config after Q2/5 bracketed 60 s. Thus 21 operating conditions, 6 scalar validation solves and 1 direct near 5-mL/min velocity solve. Previous data are preserved. Recomputed Q1 t90 differs from prior baselines by at most 0 s. Maximum relative global mass residual is 9.12e-12; concentration spans 0-1, without display-driven clipping.

Check	t90 relative change
upper_flow_grid	+2.070%
upper_flow_timestep	-0.557%
lower_flow_timestep	-0.138%
upper_flow_direct_field	+0.084%
selected_point_grid	+1.968%
selected_point_timestep	-0.304%

The direct velocity run reaches Q=4.9708 mL/min. At the same 0.3-mm grid and τ0.55, its normalized field differs from the near 1-mL/min run by 1.650% in velocity L2 and 0.294% in near-liquid P95. This quantifies inertia/boundary sensitivity, not continuum grid independence. A separately stored scalar history uses the direct field after normalization to the same comparison Q.

One-pump operation

Use one regulated downstream suction pump, with actual flow measured or calibrated through the complete assembled path and suitable inlet filtration. A sealed steady series path has the same mass flow at inlet and outlet; at equal pressure/temperature under this nearly incompressible model, Qin□Qout. Independent inlet and outlet pumps cannot impose unequal steady flows without pressure storage, leaks or compensating flow, none of which is represented here. At the Q3 candidate, one pump draws about 3 mL/min and both port rates refer to the same pressure/temperature basis. This simulation supplies no pump driver or flow-control loop; a UI record of a flow setting is not proof that physical flow was reached.

Tabulated pressure drop covers only the modeled chamber and short port segments, not filters or external tubing; it cannot alone specify a pump. Do not apply a pump's maximum vacuum to sealed culture wells. First use a cell-free water surrogate at low flow to confirm an open outlet, no visible liquid displacement and no leaks, then measure flow with all intended filters/tubes installed. This is a mechanical flow check, not a cell or sterility validation.

Reproduction and interpretation

python operating_sweep.py --run --checks --refine --selected-checks computes the cases. python operating_figures.py regenerates plots, offline HTML and this report; node test_operating.cjs tests data and interaction mathematics. In operating_sweep_3d.html, targets can be explored without changing the report's prespecified selection. X=Q, Y=D, Z=a computed time response; the third axis is not an independent control. No PCA/UMAP cluster is used to claim an optimum. Separate bilingual notes are in figure_notes.

Airflow design analysis

Decision and geometry

The selected engineering prototype is the wide inlet shelf (shelf_w8_e0p2), integrated into the left gas-cap wall. The shelf adds no component or fastener; the separate compression plate required for sealing is included in the eight-part assembly. It is a practical trade-off, not a global optimum or a biologically validated protective device.

Shelf solid: X □7.4 to □0.2 mm, Y □4 to +4 mm, Z 58.2 to 59.4 mm. Common plenum X±7.4, Y±6.0, Z 57.3-62.5 mm. Both horizontal ports: ID 2.4 mm, Y=0, Z=60.8, end X=±18 mm. The local cap roof is 1.5 mm thick to Z64, and the outer lid panel continues to Z66; the CFD inner roof remains Z62.5. A removable compression plate occupies Z56.1-57.3 with four 4×2.8 mm openings; the upper perimeter gasket spans Z57.3-57.9 while the center remains gas. The cap lower edge is 57.9. This sealing design requires eight operating prints. Four liquid cavities center (±3.3,±2.65), cross-section 4×2.8 mm, floor Z43, rim Z55.5. A 100 µL fill gives surface Z51.9286 mm. Shared gas; separate liquids.

What was compared

The objective is to avoid concentrated flow directed toward the liquid, not to create turbulence. Port Re□0.589 at 1 mL/min; directional redistribution is laminar. Eighteen actual CFD geometries cover an open plenum, an adapted 12-hole plate, nine vertical walls, three L-shaped hoods, and four shelves. Vertical walls redirected part of the inlet flow downward and increased the worst-well downward-momentum proxy. The wide shelf reduced this proxy on all three grids while remaining open and accessible for cleaning.

Results and trade-offs

The directional metric is the maximum across wells of $\rho\Sigma\max(\square u_z,0)^2\Delta A$ at a plane 1 mm below the rim. Near-liquid gas speed P95 is evaluated 0.6-1.4 mm above the fixed liquid surface. Neither is stress on submerged cells.

At 1 mL/min, finest 0.2 mm grid	Open plenum	Selected shelf
Worst-well downward-momentum proxy, N	1.058e-17	6.767e-18
Worst-well near-liquid P95 speed, mm/s	0.000970	0.000775
Internal pressure drop, Pa	0.008537	0.008607
CV of downward gas-circulation flux across wells	0.0098	0.3691

The momentum reduction is 30.9%, 30.1%, and 36.1% on 0.4, 0.3, and 0.2 mm grids, respectively. All three grids support the direction of improvement, while absolute local quantities remain grid sensitive. Near-liquid P95 decreases by 20.1% on the finest grid; this is not evidence of cell protection or uniform exposure. Gas-circulation symmetry worsens. This CV is not an odor-concentration CV; the velocity metric alone does not establish concentration; see the added gas-scalar analysis below. The perforated plate reduces near-liquid speed more on the coarse grid, and reduces mouth momentum, but creates a less accessible upper cavity. The selected design therefore includes an explicit cleanability/part-count trade-off.

Numerical method and checks

Custom double-precision 3D D3Q19 BGK lattice-Boltzmann implementation; screening τ=0.8, air ρ=1.2 kg/m³ and μ=1.8e□5 Pa·s. Halfway bounce-back on rigid walls and stationary liquid surfaces; non-equilibrium-extrapolation pressure terminals. Convergence requires relative velocity change over 200 steps <2e□5 and inlet/outlet mass imbalance <0.2%, for three consecutive reports. All 28 solves meet these declared criteria; maximum final imbalance is 0.0444%. C# versus NumPy tube-field relative L2 difference is 3.09e-12. Hagen-Poiseuille tube-flow errors are 2.54% at τ=0.8 and -1.13% at τ=0.55. Numerical benchmarks are not experimental device validation.

Screening fields were solved at small pressure forcing and Stokes-normalized to 1 mL/min. Separate τ=0.55 runs actually produced 0.9400 mL/min (open) and 0.9401 mL/min (shelf); shelf Re=0.554, relative density range 0.234%. Their normalized momentum ratio is 0.641, supporting the same directional trend near the intended operating flow. Changing τ also changes staircase-boundary error, so this is not an isolated inertia study.

Local metrics are not grid independent. The original velocity solver did not include species transport; the added gas-scalar model below still omits liquid dissolution, evaporation, moving liquid, cell injury, external filter or tubing is modeled. Internal pressure drop must not be used alone to size a pump. There is no prediction of sterility, cell viability, receptor activation or uniform exposure.

The finest-grid gas volume is approximately 1.193 mL. At 1 mL/min, V/Q is 71.6 seconds. This is only a bulk volume/flow scale, not a well-concentration equilibration time or a complete biological measurement duration. Keep the 2-second acquisition window distinct from the full odor-exposure duration.

Figures, video and reproduction

figures/design_comparison.png: 18-case screening. figures/velocity_sections.png: inlet and well sections. figures/mesh_and_tradeoff.png: grid sensitivity. airflow_comparison.mp4: three-case RK2 tracer visualization from saved steady velocity fields; 60 physical seconds in 20 playback seconds. Four hundred markers begin in the inlet and gas spaces; exited markers are reinjected. A-C show steady velocity tracers; D separately plots gas advection-diffusion concentration. No liquid/cell motion or turbulence is depicted. The raw JSON/NPZ/convergence CSV files are in results/; geometry, selection and verification are machine-readable.

Short experimental check

First use water-only wells to check leakage and establish external flow. Do not interpret a dyed-liquid flow experiment as equivalent to this gas CFD. Compare the selected cap and an otherwise identical no-shelf control at the same flow, liquid volume, temperature and cell batch. Record baseline stability and per-well response timing/amplitude. Without local gas-concentration or velocity instrumentation, describe this as a whole-system exposure comparison, not a validated velocity-field measurement. Detecting a large inter-well asymmetry is a reasonable two-week priority; elaborate flow visualization is unnecessary.

Method references

Guo, Zheng & Shi (2002), non-equilibrium extrapolation: https://cpb.iphy.ac.cn/en/article/doi/10.1088/1009-1963/11/4/310 . Nash et al., boundary-condition and curved-wall accuracy comparison: https://arxiv.org/abs/1211.0205 . The implementation is custom; citing these methods does not imply those authors validated this code.

Added: perforated sweep and gas species delivery

The original28 velocity solves are intact. Eighteen actual hole geometries plus two refined cases vary diameter1.2/1.8/2.4mm,pitch3.6/4.2/4.8mm andplate-bottomZ57.9/58.2mm. Six holes are fixed,so pitch is not global hole-count density. Minimum coarseJ/open0.0246 becomes0.0351 at0.2mm;maximum coarse ratio is2.489. Reduced circulation is not proven better cell exposure. The cleanable shelfCAD remains selected;no additional print option is released.

At100µL,liquiddepth is8.93mm and actual sub-rim gasheadspace3.57mm. Selected-shelf near-liquidP95 speed is about0.00049-0.00093mm/s on0.4mm cells. LocalPe atD10mm²/s is0.00018-0.00033,so diffusion can supply gas despite tiny tracer displacements. L²/D across the gasgap is1.276s;whole inlet/plenum delivery givesNE near-liquidt90=149.55s,versus152.09s on0.3mm cells. Delivery through liquid to bottom cells is distinct;see the adjacent membrane analysis.

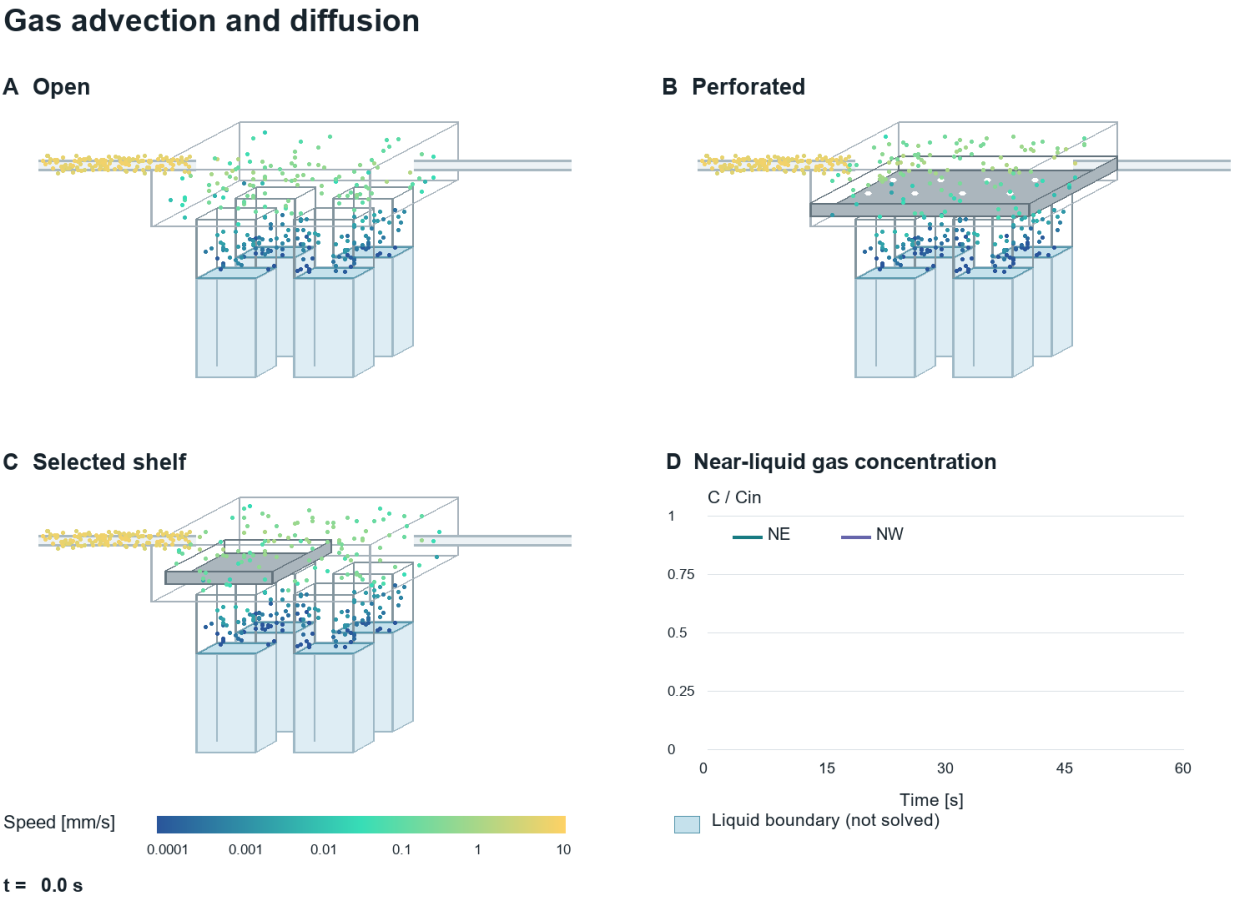
The gas scalar uses conservative cell-centredFV,backwardEuler,upwind advection andcentral diffusion. Interpolated face fluxes are projected to discrete incompressibility;baseline correctionL2 is1.54%. Inlet concentration steps to1 from initial0;outlet isconvective with no diffusiveflux;walls andliquid areimpermeable. D5/10/20mm²/s are unmeasured VOC sensitivity assumptions. Maximum relative mass-ledger error8.65e-12;1D analytic mean-concentration error0.000246;baseline t90 changes+1.70% ongridrefinement and□0.20% with halveddt. These are numerical checks,not fullgrid independence or physical validation.

Paper-style figures have separate bilingual notes indexed byFigure_Guide. Every offline3D point comes fromhole_sweep.csv. Browserinteraction was not tested;source/data/rotation-math checks are reported separately. Gas concentration does not establish liquid uptake,PDMS transfer,cellresponse,filters or externaltubing behaviour.

Method background:NIST FiPy conservation equation,cell-centred finitevolume. This customimplementation does not useFiPy or implyNIST validation.

For the added fixed-STL flow-rate selection, see Gas operating conditions (Flow_Operation_EN.md). The1-mL/min results above remain the baseline.

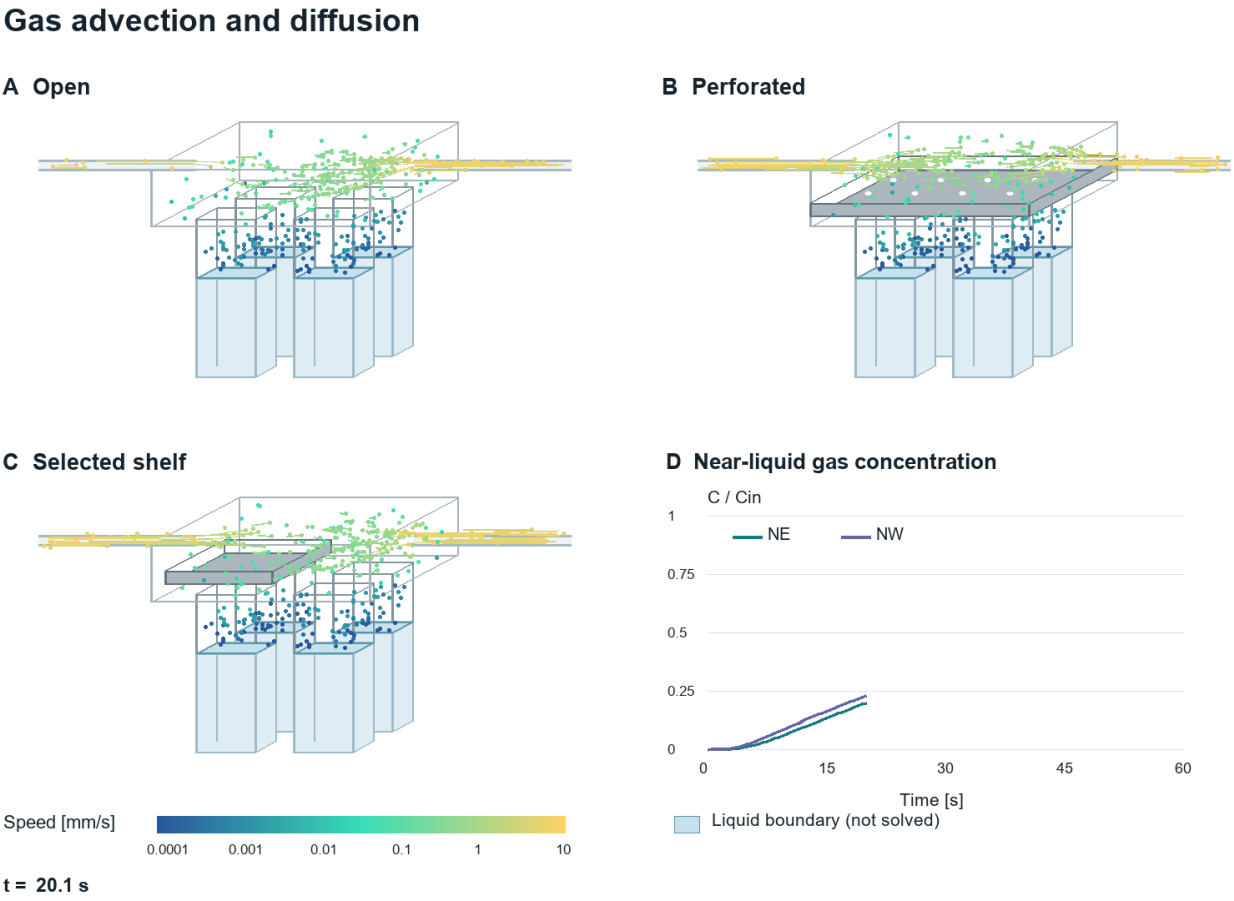
Airflow and gas transport / airflow_movie_000



Movie | A-C areRK2 tracers in saved steady velocity fields;D independently shows selected-shelf gas concentration atDgas10.60physical seconds play in20seconds. Poster at physicalt□0.0s.

02_simulation\airflow\figures\airflow_movie_000.png

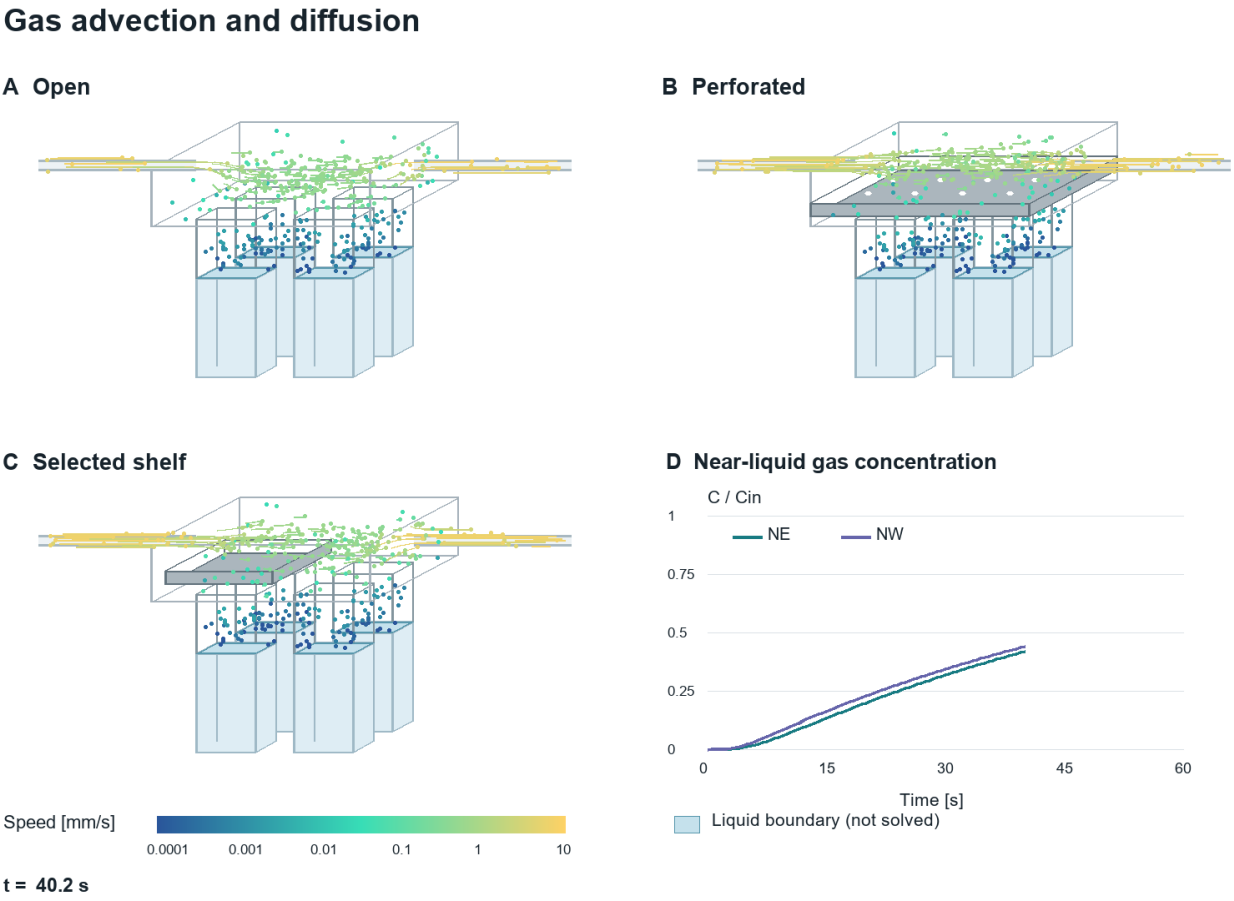
Airflow and gas transport / airflow_movie_080



Movie | A-C areRK2 tracers in saved steady velocity fields;D independently shows selected-shelf gas concentration atDgas10.60physical seconds play in20seconds. Poster at physicalt□20.1s.

02_simulation\airflow\figures\airflow_movie_080.png

Airflow and gas transport / airflow_movie_160



Movie | A-C areRK2 tracers in saved steady velocity fields;D independently shows selected-shelf gas concentration atDgas10.60physical seconds play in20seconds. Poster at physicalt□40.2s.

02_simulation\airflow\figures\airflow_movie_160.png

Airflow and gas transport / design_comparison

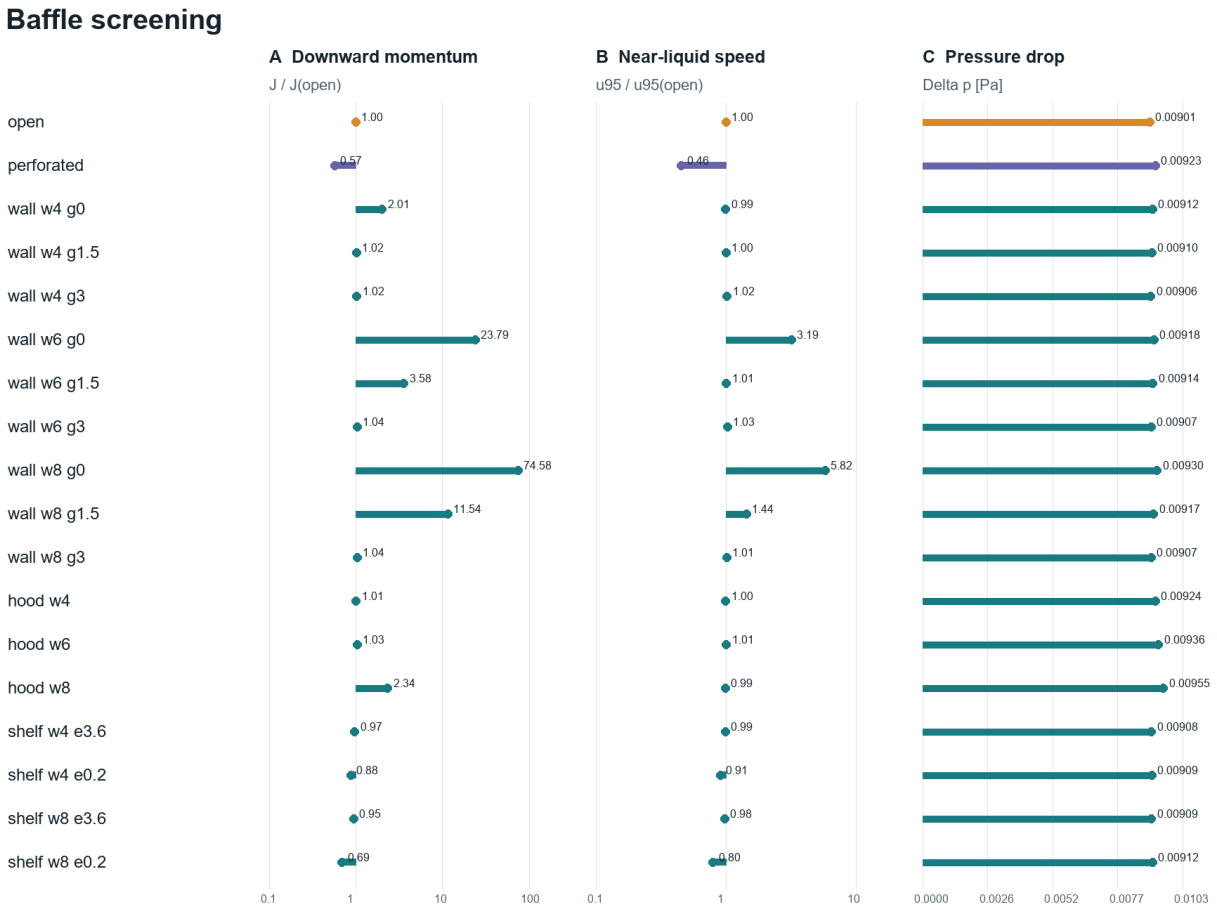


Figure1 | Original18-geometry baffle screening. A: worst-well downward momentum and B: near-liquid P95 speed normalized to the open plenum; C: internal pressure drop.

02_simulation\airflow\figures\design_comparison.png

Airflow and gas transport / gas_delivery

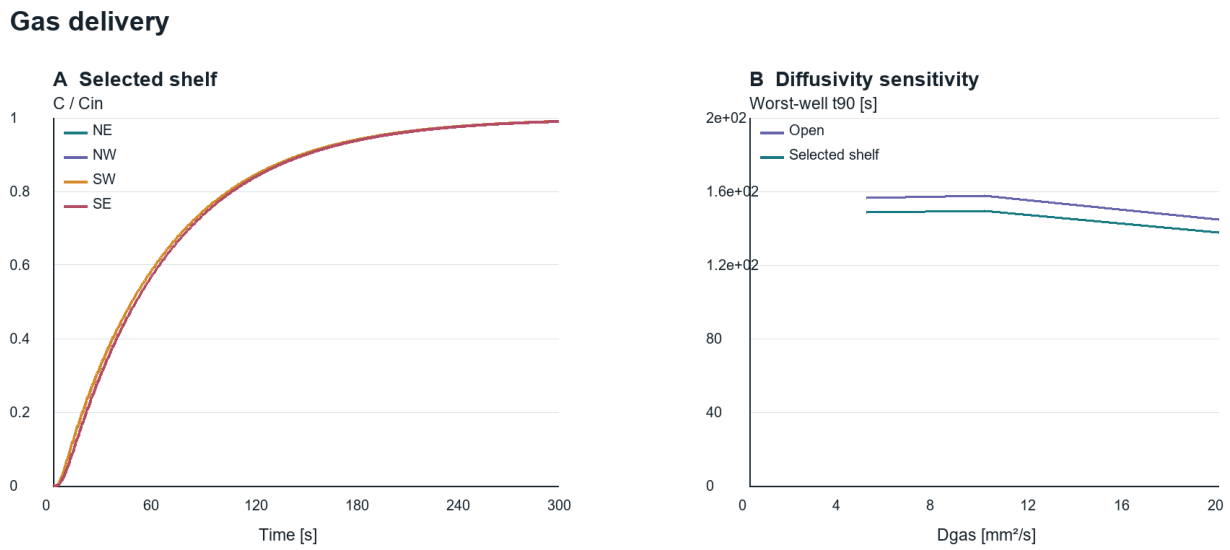


Figure5 | Gas species can reach a well by diffusion while velocity tracers barely move.

02_simulation\airflow\figures\gas_delivery.png

Airflow and gas transport / hole_sweep

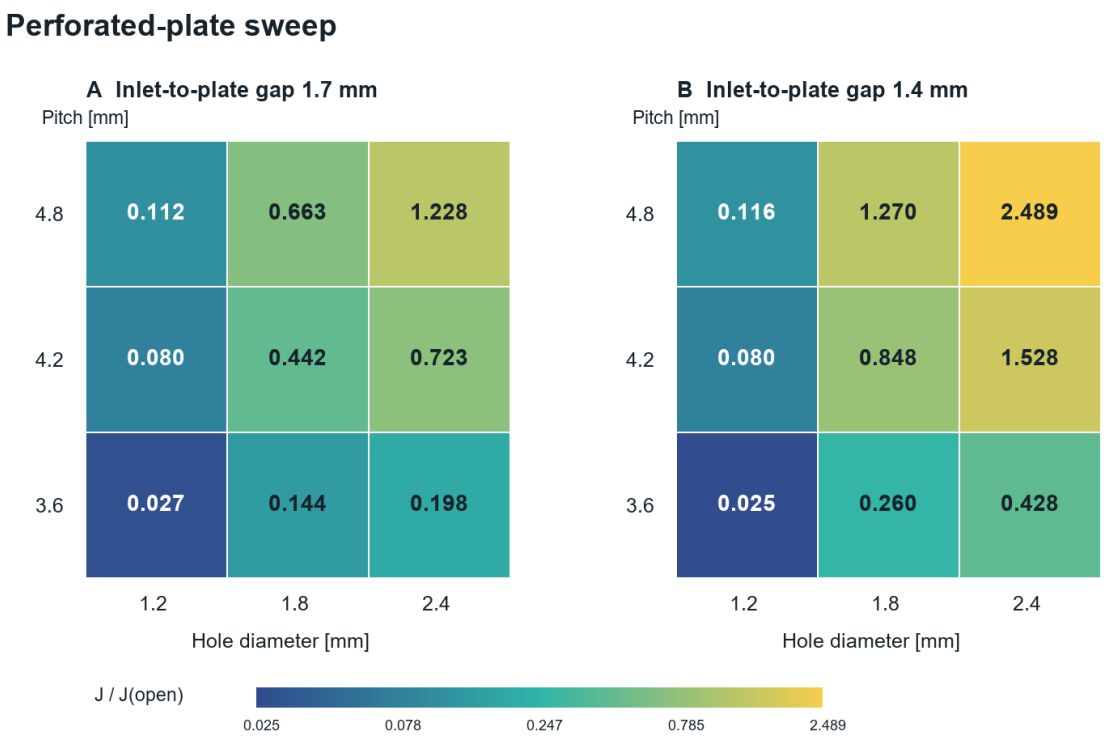


Figure4 | Actual18-point perforated-plate factorial sweep; small holes do not by themselves establish a better device.

02_simulation\airflow\figures\hole_sweep.png

Airflow and gas transport / mesh_and_tradeoff

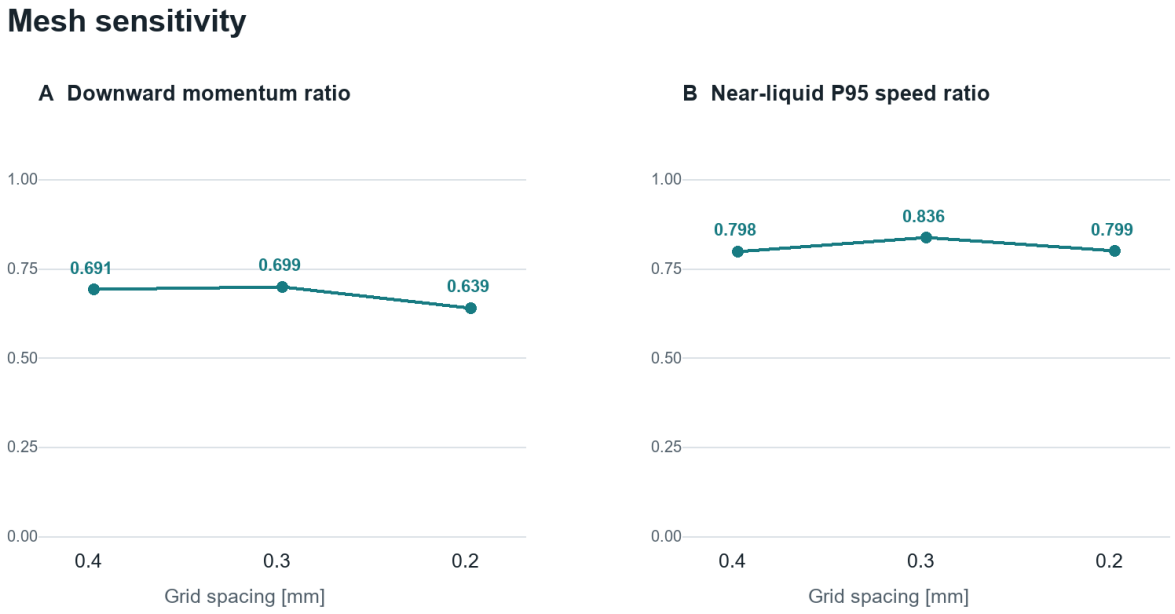
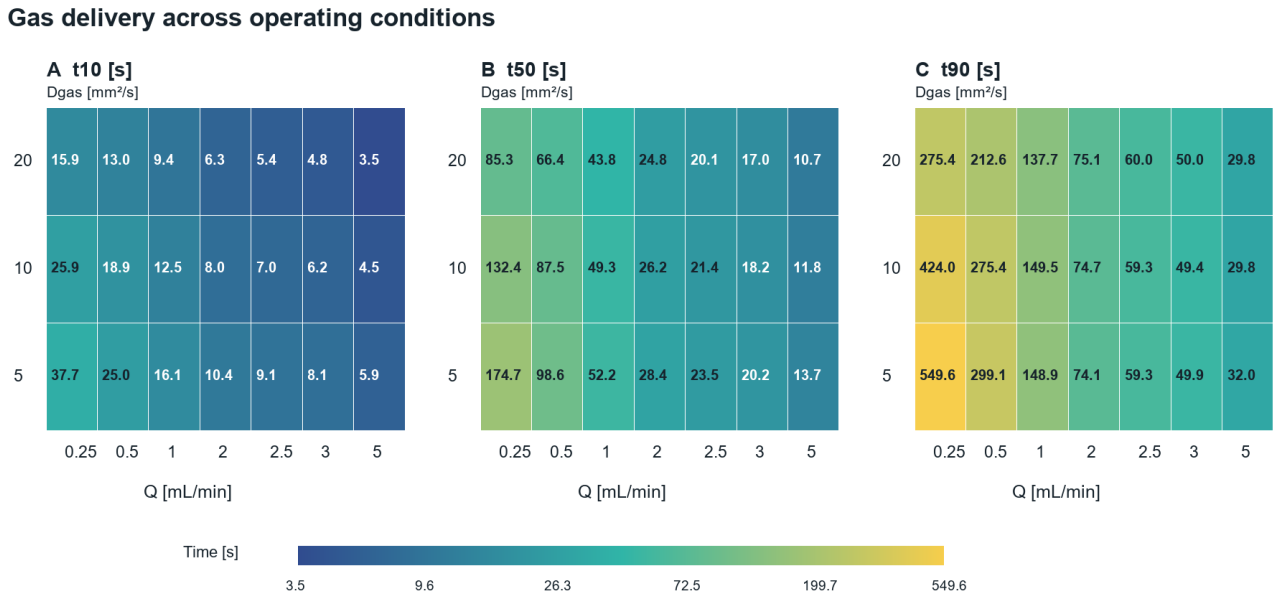


Figure2 | Mesh sensitivity of the selected shelf relative to the open plenum.

02_simulation\airflow\figures\mesh_and_tradeoff.png

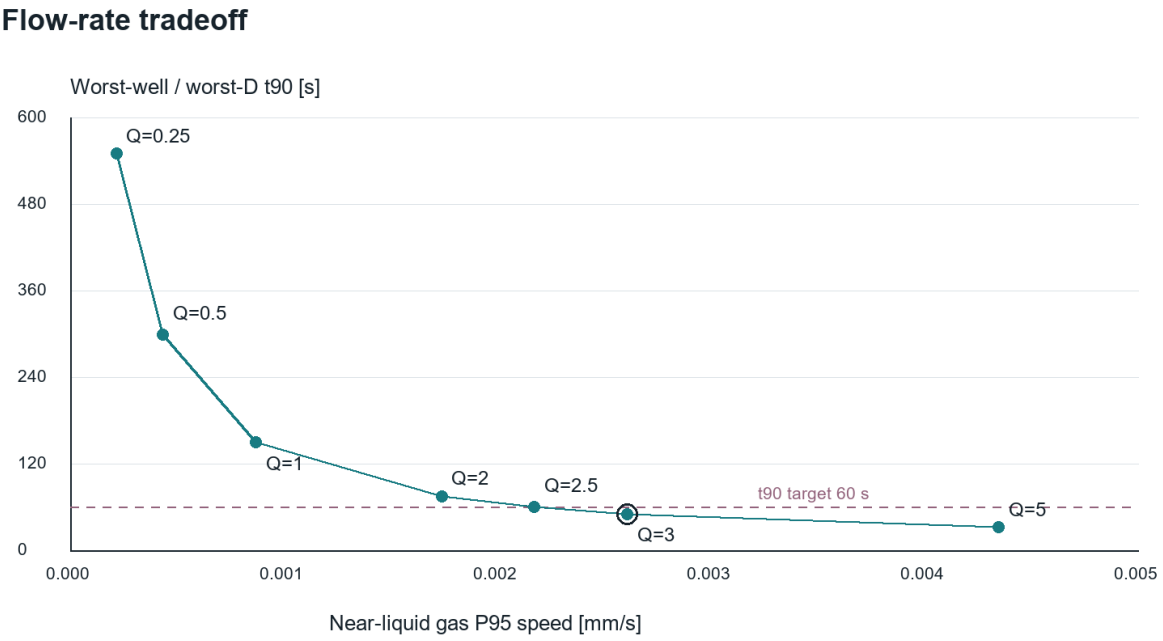
Airflow and gas transport / operating_delivery



Flow-diffusivity sweep for the fixed shelf; each cell is the slowest-well gas t10, t50 or t90.

02_simulation\airflow\figures\operating_delivery.png

Airflow and gas transport / operating_pareto



Nondominated tradeoff between gas delivery time and near-liquid speed; the ring marks the lowest sampled flow meeting the provisional constraints.

02_simulation\airflow\figures\operating_pareto.png

Airflow and gas transport / velocity_sections

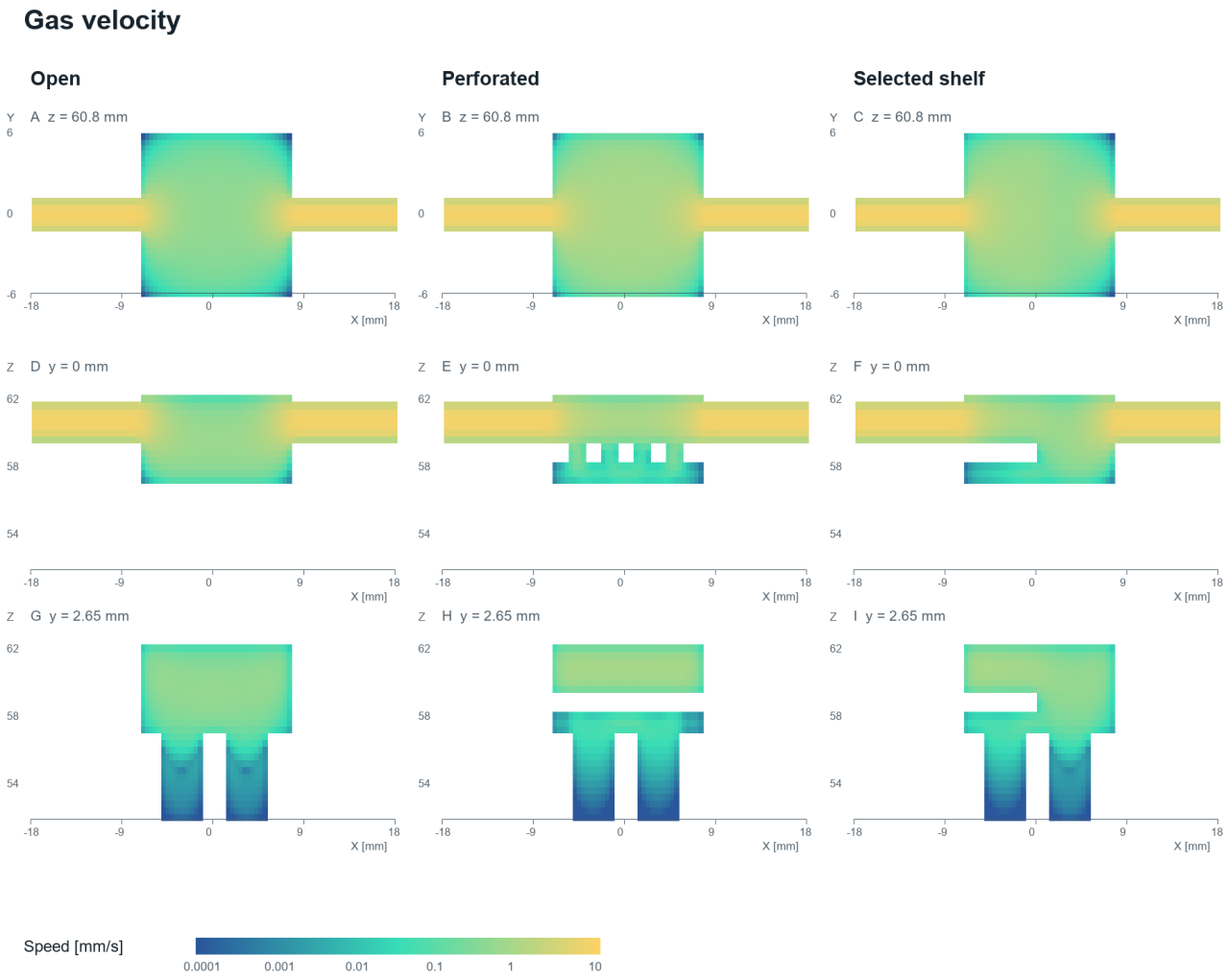


Figure3 | Velocity sections for open, original perforated and selected-shelf cases.

02_simulation\airflow\figures\velocity_sections.png

Molecular delivery and the proposed PDMS film

This is a one-dimensional transport screening study, separate from the three-dimensional gas-flow CFD and from biological response. The confirmed operating geometry is 100 µL in a 4 × 2.8 mm well, with HEK293T cells attached to the bottom. Thus the nominal liquid column is 8.929 mm and the gas space below the rim is 3.571 mm. The calculations do not establish actual cell exposure, viability, receptor response or GCaMP response time.

Decision for the current build

Keep the current eight-part mechanical set as the experimental carrier. Do not add PDMS to the selected build or mandatory BOM yet. Neither the target VOCs nor their membrane/medium transport properties are established. The current carrier is not thereby validated for rapid gas-to-cell delivery: the deep liquid column is an unresolved transport constraint that should be measured before making a rapid-response claim. Thicknesses in this study are a research grid, not alternative print files.

PDMS can be useful as a selective permeation layer under a maintained concentration gradient. It is not a universal VOC-only gate. A membrane that admits a compound still allows it to equilibrate in a closed receiver; without a sink, this model's final downstream gas-equivalent concentration equals upstream concentration. Selectivity primarily affects delivery rate, transient composition and retention, not permanent rejection at equilibrium. The molecular target, interferents, humidity, acceptable delay and recovery criterion must therefore be defined first.

What the literature supports

- Hydrophobicity describes wetting by liquid water. It does not establish low water-vapor transmission. Bian et al. measured water-vapor transport through 8-160 µm cross-linked PDMS films; the thinnest samples also showed that surface kinetics can invalidate simple inverse-thickness extrapolation. This material was not a certified match to the team's formulation. Primary study

- Heo et al. measured evaporation and medium-osmolality changes through PDMS, including humidified incubation. Their conditions and cell systems differ from AeroSense; the paper supports checking water loss, not predicting AeroSense viability from it. Primary study

- PDMS admits oxygen and carbon dioxide. Firpo et al. report, at 293 K, oxygen permeability 2.8 × 10□¹³ mol m□¹ s□¹ Pa□¹ and diffusivity 3.1 × 10□□ m²/s. Their measured surface resistance also cautions against treating thin films as bulk-only barriers. These are reference data, not measurements of the proposed film. Primary study, Table 1

- Small molecules can partition into PDMS, and solvent exposure can cause swelling. Retained analyte and later desorption can alter delivered pulses and create carryover; the magnitude is species- and condition-dependent. Toepke and Beebe, Lee, Park and Whitesides

- Lue et al. measured toluene/nitrogen mixtures with two cross-linker formulations and concentration-dependent transport. This does not supply a universal 5:1 → 10:1 → 20:1 permeability law. We assign no diffusivity, permeability, selectivity or optimum to those ratios. Primary study

Model and boundaries

For each layer, physical concentration is c = K u, where u is gas-equivalent concentration and K = c_layer/c_gas at equilibrium, using the same mol/m³ concentration units. Gas has K = 1. The equation is K ∂u/∂t = ∂(D K ∂u/∂x)/∂x. Interface fluxes conserve mass and use the sum of both half-cell diffusion resistances. Optional surface resistance adds 1/h at each PDMS interface; h uses gas-equivalent concentration units, m/s.

The ordered layers are optional PDMS at the well mouth, gas headspace, and liquid. Sidewalls are impermeable; the bottom has zero flux, representing an initially unloaded, non-consuming transport probe at the cell position. Initial concentration is zero and the upstream concentration steps to one. The reported bottom fraction is c_liquid/(K_liquid c_gas,input). Instantaneous partition equilibrium, dilute linear sorption, constant D and K, uniform cross section, no liquid convection, no binding, no chemical reaction and no cell consumption are assumed. A real surface meniscus, PDMS deflection, leaks and bypass around a film are excluded.

The default input is a step immediately above the well rim. It omits upstream tubing, filters and plenum wash-in and is therefore favorable with respect to arrival delay. A separate sensitivity replaces it with a perfectly mixed 0.8544 mL plenum supplied at 0.1, 1 or 10 mL/min, with four identical wells drawing from that reservoir. That approximate volume is the rectangular plenum minus the shelf; ports are excluded. This is a reduced model, not the CFD flow field. couple_gas_history.py accepts a measured/CFD rim history; it is one-way forcing and does not recalculate gas transport after absorption changes the boundary.

The finite-volume system is transformed to a symmetric operator and propagated by its matrix exponential. There is no finite time-step error for step forcing. The last liquid-cell value approximates the impermeable bottom and is checked by spatial refinement. forced() integrates a piecewise-linear upstream history analytically in time. All governing units are SI; plotted hours and µm are converted from seconds and metres.

Inputs: measured geometry versus assumed properties

Input	Values	Status
Well area / operating fill	11.2 mm² / 100 µL	Current geometry and user-confirmed fill
PDMS thickness	25, 50, 100, 250, 500 µm; plus no film	Hypothetical research grid, not a chosen membrane
VOC diffusivity in PDMS	10□¹², 10□¹¹, 10□¹□, 10□□ m²/s	Broad sensitivity bounds; no species assignment
PDMS/gas partition K	1, 10, 100, 1000	Assumed dimensionless concentration ratios
Liquid/gas partition K	0.1, 10, 1000	Assumed; not a measured Henry coefficient
Molecular liquid D	0.5, 1, 2 × 10□□ m²/s	Illustrative sensitivity, not a target-VOC measurement
Gas D	10□□ m²/s	Screening assumption; zero convection in the 1D gas layer
Effective liquid D multiplier	1, 10, 100	Mixing sensitivity proxy, not a solved mixing flow
Surface h	Ideal equilibrium or 10□□-10□³ m/s	Assumed kinetic-resistance sensitivity
Horizon	7 days	Numerical censoring limit; not a permitted cell incubation

The generic VOC sweep is isothermal with no assigned chemical species or fitted temperature law; its D/K values are independent assumed inputs. Changing temperature requires corresponding measured properties. Only the separate oxygen example explicitly uses 293 K reference data. A water drying rate is not calculated: PDMS-water partitioning/permeability, relative humidity, temperature and external transfer resistance remain unknown.

Main calculation results

For an ideal step fixed at the liquid surface and a stagnant, nonreacting liquid with an impermeable bottom, the exact series solution gives t10 = 0.130159 L²/D, t50 = 0.378748 L²/D and t90 = 1.031105 L²/D. These are concentration arrival times, not receptor/GCaMP timing.

Assumed liquid D (m²/s)	t10	t50	t90
0.5 × 10□□	5.76 h	16.77 h	45.67 h
1 × 10□□	2.88 h	8.39 h	22.83 h
2 × 10□□	1.44 h	4.19 h	11.42 h

This is a serious conditional bottleneck, not a measurement that the actual device must take 23 hours. Convection, mixing, binding and species-specific solubility may change delivery. Faster gas flow above the well is not evidence that the liquid column has mixed. In the same analytical benchmark, the 3.571 mm gas layer at assumed D = 10□□ m²/s has t90 □ 1.315 s despite zero velocity: molecular diffusion remains active

when advective tracers appear stationary.

At fixed cross section and D, the liquid diffusion time scales as fill volume squared. The 25 µL counterfactual gives t90 □ 1.43 h at D = 10□□ m²/s. It is a sensitivity result, not permission to lower the confirmed operating fill or an automatically selected new well. At 100 µL, multiplying effective liquid D by 100 reduces the no-film screening t90 to about 0.25 h; adding a nominal 100 µm PDMS film raises it to about 0.85 h. A membrane can become the limiting resistance once liquid transport is faster.

The full grid contains 243 conditions, including three no-film baselines. Seventy-four do not reach t90 within seven days; their CSV time is blank and the figure marks >168 h. Do not replace these with zero. Representative 100 µL, D_liquid = 10□□, K_liquid = 10 results are: no film t90 □ 22.85 h; 100 µm PDMS with assumed D = 10□¹□ and K = 100, t90 □ 23.39 h; the same film with a 1 mL/min finite-feed reservoir, □ 23.55 h. These examples identify sensitivities and do not identify an optimal commercial PDMS.

Intrinsic concentration permeability is P_c = D K in m²/s; bulk permeance is D K/L in m/s. Pressure-based permeability is P_p = D K/(R T) in mol m□¹ s□¹ Pa□¹. They must not be interchanged numerically. Intrinsic A/B flux selectivity under comparable maintained gradients is (D_A K_A)/(D_B K_B). Film thickness changes both permeances together in this bulk model and does not itself change that intrinsic selectivity. Surface kinetics, mixed sorption and swelling can break this simple result.

The bulk time lag L²/(6D) is not the bottom-cell t90. For 100 µm and D = 10□¹□ it is 16.67 s, while the liquid-column response remains hours. A 20-minute input pulse can also produce a delayed, broad bottom concentration peak after input has stopped. In the no-film example its peak is approximately 2.78% of the input equilibrium reference near 3.92 h. The plotted storage/pulse curves demonstrate reversible diffusion memory only; irreversible adsorption and receptor adaptation are not modeled. Increasing K increases both storage and permeability, so it does not universally increase response delay.

Executed one-way chain from the gas CFD

integrate_airflow.py propagates the selected shelf case's NE-rim history (Q = 1 mL/min, assumed gas D = 10 mm²/s) through the 100 µL no-film and nominal 100 µm examples. The source CFD ends at 300 s; its rim t50 is 48.78 s and t90 is 149.02 s. The sampled plane is Z = 55.329 mm, used as a proxy for the nominal rim at 55.5 mm. The liquid-bottom fraction remains below a numerical reporting floor of 10□¹□ in both examples at 300 s. Tiny signed raw values are floating-point cancellation, not negative physical concentration or a validated detection limit.

For a transparent long-time comparison, two explicitly assumed continuations begin at 300 s: the rim immediately reaches one, or it remains at its final CFD value 0.991144. They bound future histories only if the future rim stays between those values. Relative to the original inlet reference, no-film t90 is 22.87-23.62 h and nominal-film t90 is 23.41-24.18 h. The immediate-to-one continuation adds about 67.3 s to the respective ideal-rim-step t90. The larger held-value shift includes the permanently lower boundary concentration; it is not entirely a gas wash-in delay. No 23-hour CFD has been run or implied.

This chain preserves the 1D solver's conservation for its prescribed boundary, but it is not a mass-conserving, two-way coupled gas-membrane-liquid simulation. The upstream CFD imposes an impermeable liquid boundary and is not recalculated after liquid or PDMS uptake. The example shows that the assumed deep stagnant liquid still dominates the time scale; it does not establish actual exposure or whole-device mass balance. Input hashes, assumptions, supplied-time histories and separate continuation histories are in results/CFD_rim_coupling_summary.json and the associated CSVs.

Oxygen, carbon dioxide and water

The oxygen calculation is a capacity ceiling under assumed steady, stagnant transport. It uses the cited 293 K PDMS P/D, assumed liquid D = 2 × 10□□ m²/s, assumed surface dissolved oxygen 0.30 mol/m³ and gas oxygen partial pressure 21,278 Pa. At 100 µL, maximum supply to a zero-concentration bottom is about 0.75 pmol/s per well; a 100 µm film barely changes this example because liquid resistance dominates. The demand sweep 0.01-5 pmol/s is an assumed load, not a measured HEK293T respiration rate or a cell count. Above capacity, the zero-order stationary solution is infeasible; negative concentration is not reported as physical.

This does not establish safe oxygenation, CO2/pH balance or a 37°C culture condition. Medium composition, gas composition, cell number, respiration, mixing and target dissolved oxygen must be measured. Likewise, water-repellent PDMS can still lose water vapor. No film is promoted as a moisture barrier, sterile barrier, validated pressure seal or guaranteed analyte filter.

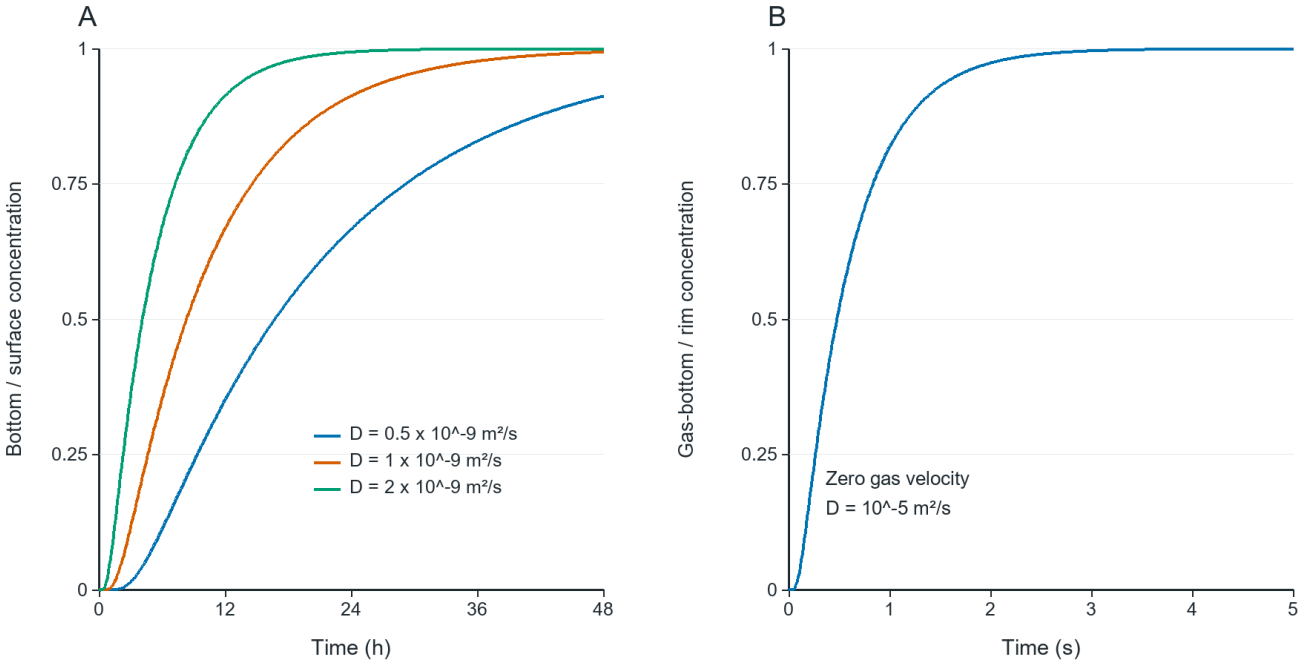
Numerical checks and the minimum next evidence

The 128-cell standalone membrane test reproduces $L^2/(6D)$ within 0.00305%; the liquid-bottom t90 error versus the exact series is 0.000762%. In the representative coupled model, increasing liquid cells from 48 to 96 changes t90 by 0.00405%. The worst normalized mass-balance residual across the broad membrane sweep is about 2.17 × 10□□; all sampled concentrations stay within numerical tolerance of [0,1]. These checks validate the implementation under its assumptions, not the assumptions themselves. Full results are in results/validation.json, membrane_sweep.csv and summary.json.

Before adding a film, define one target VOC and one relevant interferent, the gas concentration units, humidity, temperature, formulation and acceptable delay/recovery. Use the blank measurement templates to record the actual film thickness and condition. Measure a no-film baseline and a film comparison with an independent concentration assay; GCaMP alone cannot separate transport, receptor sensitivity and optical effects. Measure the actual 100 µL bottom-delivery behavior first. A calibrated cell-free tracer can test liquid mixing, but a different molecule does not establish target-VOC partitioning or membrane selectivity. These are priorities, not an instruction to perform every point in the large simulated grid.

Keep the mechanical set unchanged while those data are obtained. If rapid bottom exposure is not achieved, liquid-path redesign or verified mixing must be considered before presenting a rapid biosensor. Adding PDMS without resolving that bottleneck is not supported by these results.

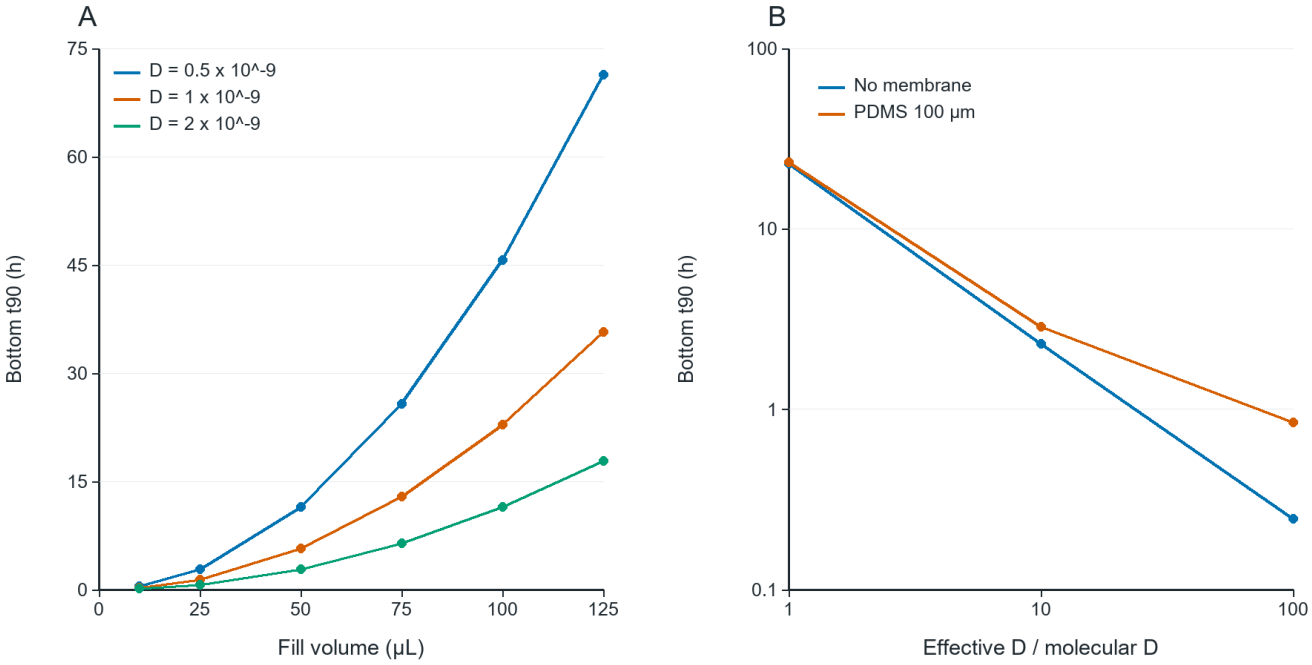
PDMS and liquid transport / 01_gas_and_liquid_times



One-dimensional diffusion steps with no membrane or reaction. (A) Exact bottom concentration for the confirmed 8.929 mm liquid depth at three assumed liquid diffusivities. The surface is fixed immediately; this is not measured cell response. (B) An isolated 3.571 mm gas layer, with its mouth instantly fixed and its bottom impermeable, has t90 approximately 1.315 s at zero velocity and assumed D=10□5 m²/s. It excludes upstream wash-in and liquid uptake; it is not the approximately 149 s inlet-to-rim CFD result. The panels have different time units.

02_simulation\membrane\figures\01_gas_and_liquid_times.png

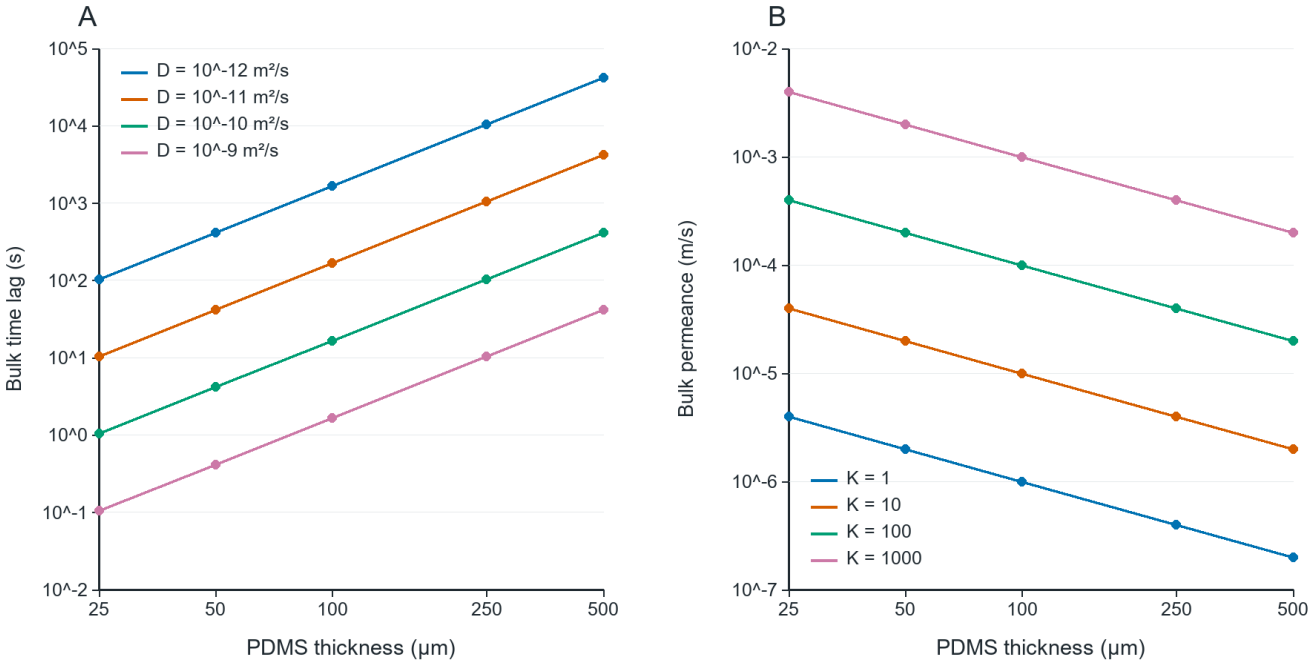
PDMS and liquid transport / 02_depth_and_mixing



Sensitivity to liquid-path assumptions. (A) Bottom t90 scales quadratically with fill volume at fixed11.2mm² area; only100µL is the confirmed operating condition. (B) Effective liquid D multipliers are mixing proxies, not computed flow. The100µm film uses assumed D_PDMS=10□10m²/s,K_PDMS=100,K_liquid=10. Membrane resistance becomes more visible when liquid transport is faster.

02_simulation\membrane\figures\02_depth_and_mixing.png

PDMS and liquid transport / 03_intrinsic_PDMS



Bulk solution-diffusion screening over25-500µm, not a selected film. (A) Time lag L²/(6D) for assumed PDMS diffusivities. This lag is not bottom-cell t90. (B) Concentration-based permeance DK/L at D=10□10m²/s and four assumed dimensionless PDMS/gas partition ratios. Surface resistance, swelling and mixed-species effects are excluded from these intrinsic curves.

02_simulation\membrane\figures\03_intrinsic_PDMS.png

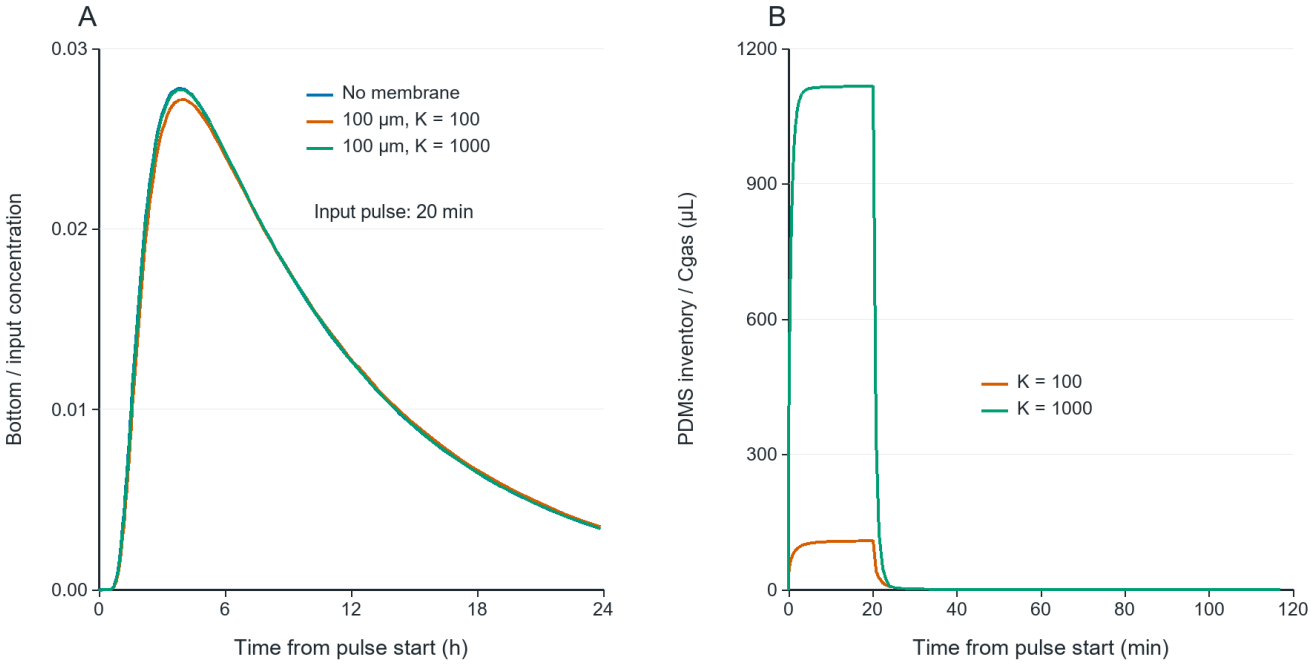
PDMS and liquid transport / 04_PDMS_coupled_sweep



Bottom t90 for a PDMS-gas-liquid stack at100µL,D_liquid=10□9m²/s,K_liquid=10. (A) K_PDMS/gas=1. (B) K_PDMS/gas=100. Rows are assumed PDMS diffusivity; columns are membrane thickness. Cell numbers are hours; gray cells did not reach90% within168h. These are illustrative inputs, not measured properties of a VOC or curing ratio.

02_simulation\membrane\figures\04_PDMS_coupled_sweep.png

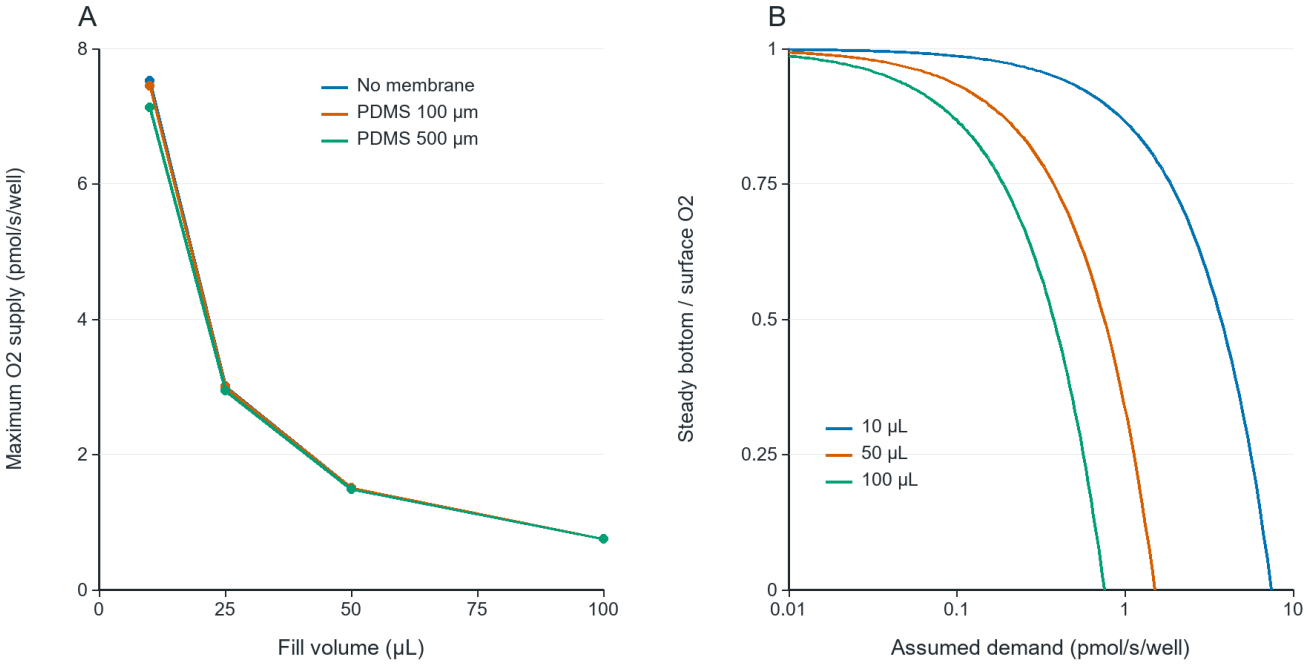
PDMS and liquid transport / 05_pulse_and_storage



A20-minute prescribed gas pulse in the representative100µL stack. (A) Bottom concentration normalized by K_liquid times input gas concentration; the broad peak occurs after the input has ended. (B) Reversible PDMS inventory divided by input gas concentration, reported as gas-equivalent volume per well. D_PDMS=10□10m²/s,D_liquid=10□9m²/s,K_liquid=10. No irreversible adsorption or receptor adaptation is modeled.

02_simulation\membrane\figures\05_pulse_and_storage.png

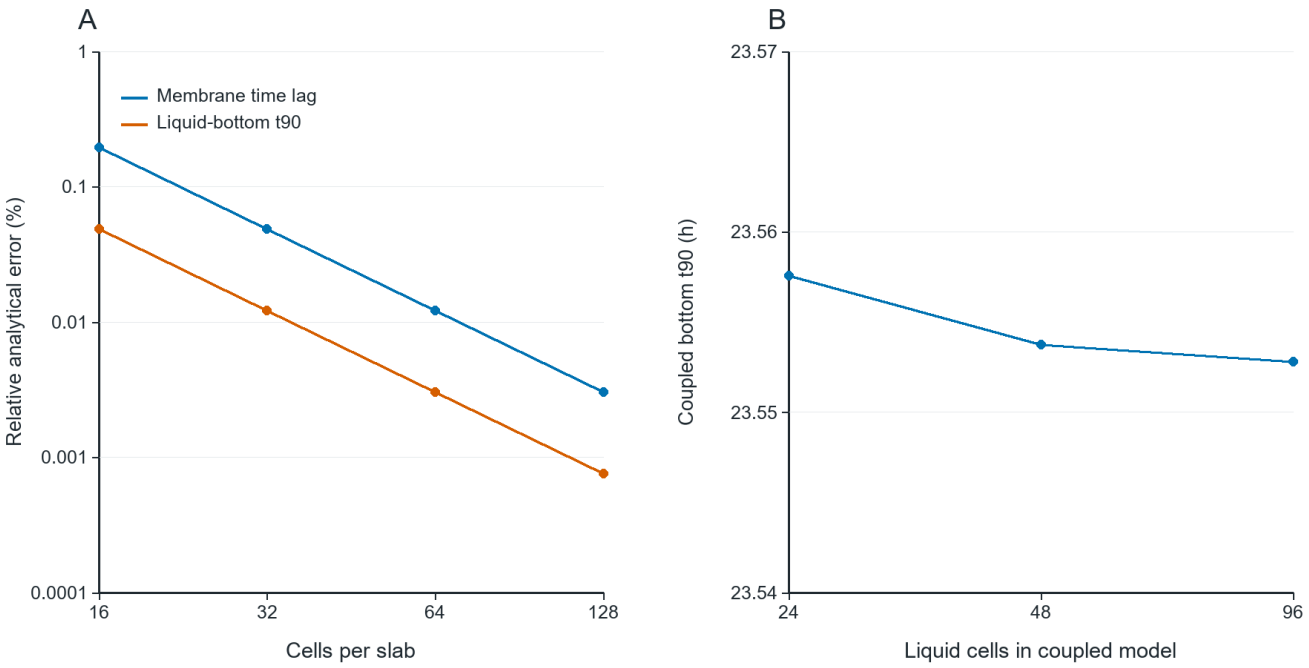
PDMS and liquid transport / 06_oxygen_capacity



Illustrative oxygen-transport capacity at293K, not a viability prediction. Published reference PDMS P/D are combined with assumed liquid D=2×10□9m²/s and surface O2=0.30mol/m³. (A) Maximum steady supply to a zero-concentration bottom. (B) Constant-demand bottom fraction for100µm PDMS; curves terminate when assumed demand reaches transport capacity. Loads are not measured HEK293T respiration rates; gas/medium control and37°C conditions are not validated.

02_simulation\membrane\figures\06_oxygen_capacity.png

PDMS and liquid transport / 07_numerical_validation



Implementation checks. (A) Spatial refinement of membrane time lag and liquid-bottom t90 against analytical solutions. (B) Representative100µm PDMS-gas-liquid model with1mL/min mixed-reservoir feed;48→96 liquid cells changes t90 by approximately0.00405%. Time evolution uses a matrix exponential. Small numerical error does not validate assumed material properties or biological performance.

02_simulation\membrane\figures\07_numerical_validation.png